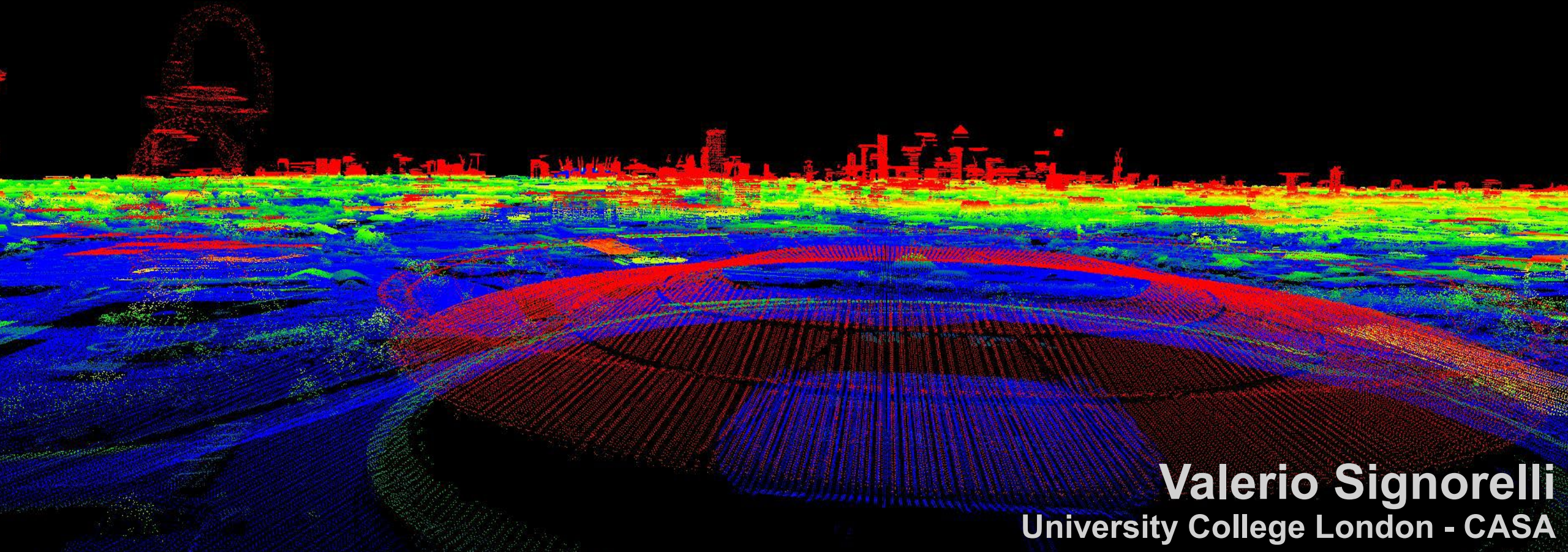


The Other Spatial Computing - Visualising Data in Context



Valerio Signorelli
University College London - CASA



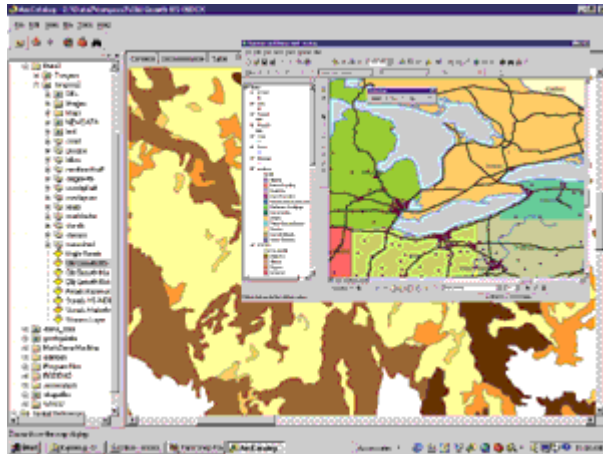
Connected Environments



Spatial Computing

- Spatial Computing date back to the '80ies. In Geographic Information Science - *computing on large-scale geospatial information, maps, and spatial datasets*
- A broad definition “**the digitalization of physical space, enabling computers to understand, map, and manipulate 3D geographic data, real-world objects, and spatial relationships**”
- In early 2000 the term started to be used in a HCI following the growing interest in immersive technologies

GIS - *Space* is geographic data to compute on



ArcInfo by ESRI

HCI - *Space* is the medium for interaction



VPS by Niantic

The other Spatial Computing

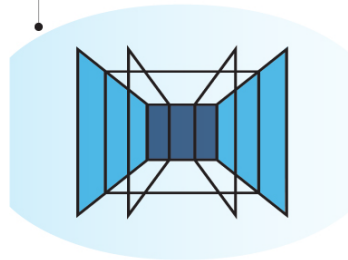
Use of space around us as a medium to interact with technology

Spatial computing refers to human interaction with a machine that can understand, retain, and manipulate references to real objects and spaces, treating the physical environment itself as part of the interface.

Greenwold (2003). Spatial computing.
Media Lab, Massachusetts Institute of Technology.

VIRTUAL REALITY (VR)

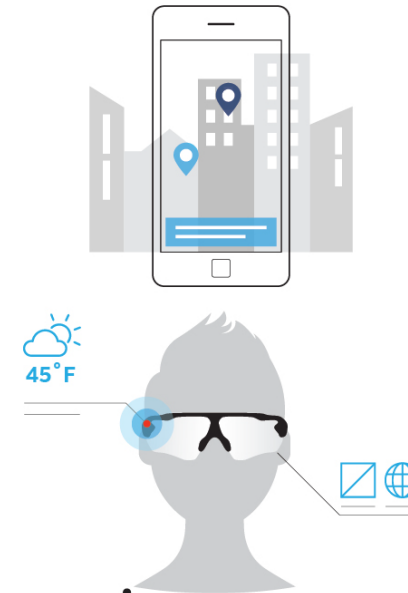
Completely digital environment



Fully enclosed, synthetic experience with no sense of the real world.

AUGMENTED REALITY (AR)

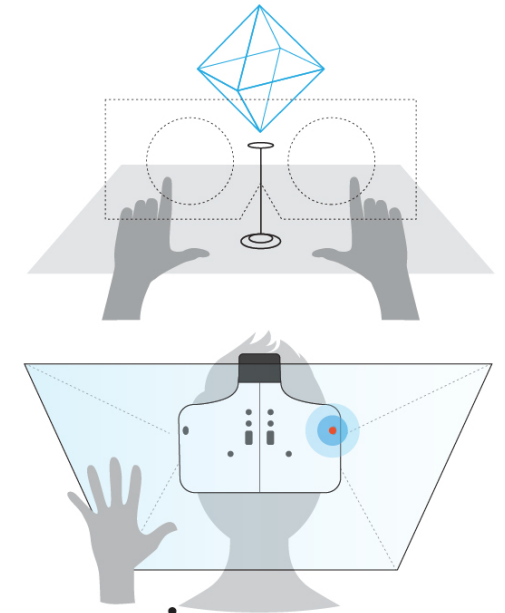
Real world with digital information overlay



Real world remains central to the experience, enhanced by virtual details.

MERGED REALITY (MR)

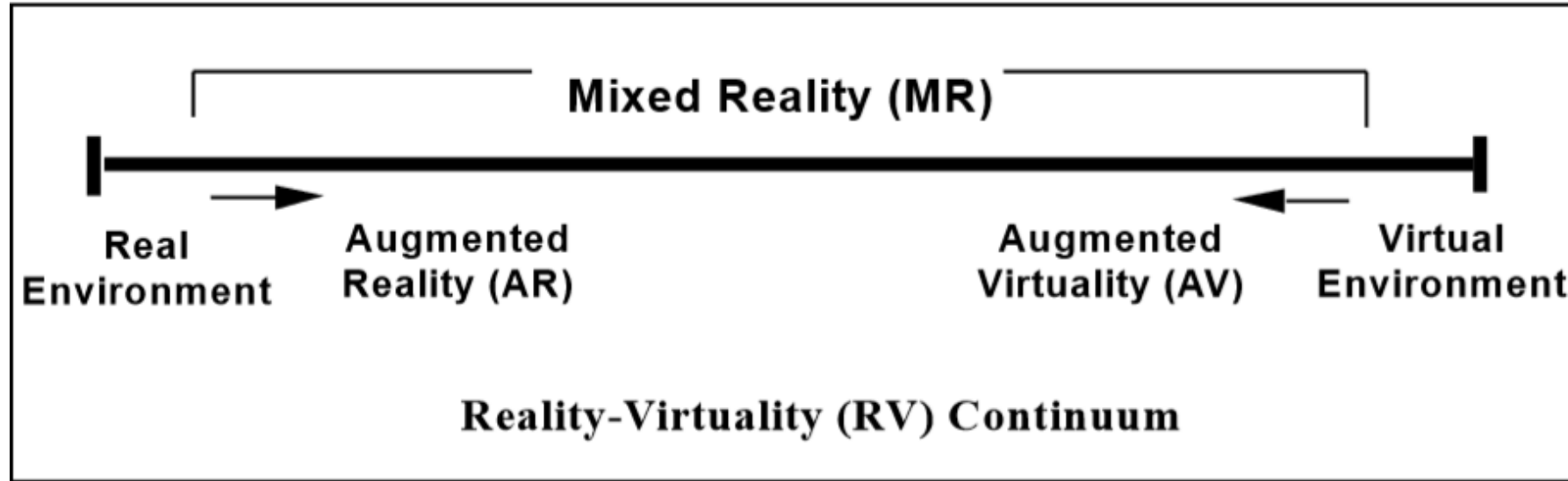
Real and the virtual are intertwined



Interaction with and manipulation of both the physical and virtual environment.

Infographic Intel Lab

The other Spatial Computing



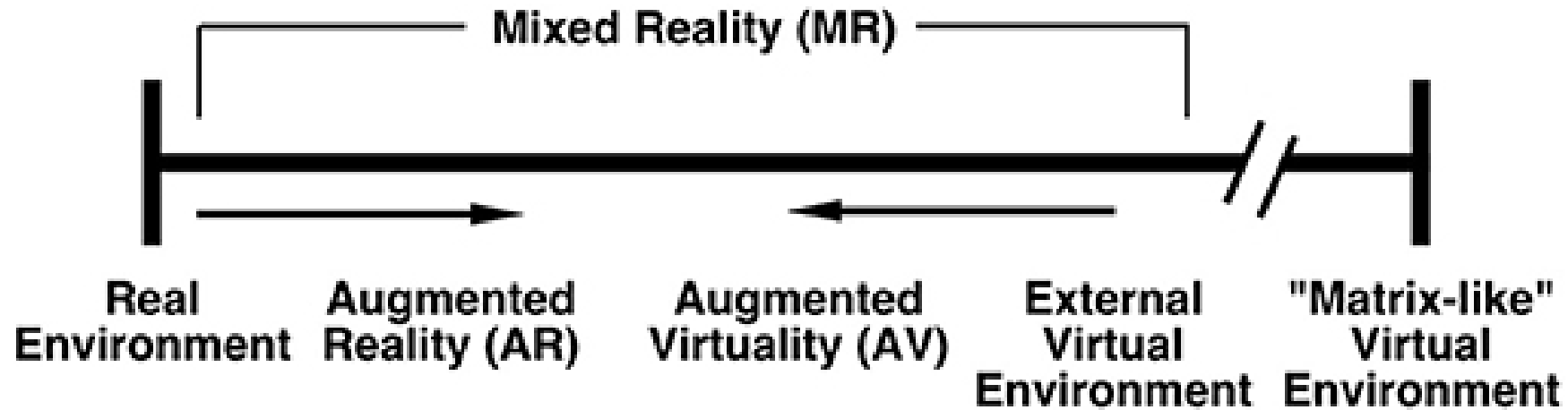
- Real Environment and Virtual Environment - totally immersed in a synthetic world
- To categorise MR displays - real world and virtual world objects are presented together
- Focused on visual display – “See-through” AR display / WoW window-on-the-world

Milgram, P., & Kishino, F. (1994). A Taxonomy of Mixed Reality Visual Displays. *IEICE Trans. Information Systems*, E77-D, no. 12, 1321–1329.

Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995). *Augmented reality: A class of displays on the reality-virtuality continuum* (H. Das, Ed.; pp. 282–292). <https://doi.org/10.1117/12.197321>

Is it still valid today?

The other Spatial Computing



- **Not a continuum:** “perfect VR” is not achievable (for now)
- **Not displays, but experiences:** focus on mixed-reality experiences, not devices
- **MRE definition:** real and virtual objects/stimuli presented together within a single percept
- **Holistic view:** mediating technology, conveyed content, and user experience considered together

Skarbez, R., Smith, M., & Whitton, M. C. (2021). Revisiting Milgram and Kishino's Reality-Virtuality Continuum. *Frontiers in Virtual Reality*, 2.

<https://www.frontiersin.org/articles/10.3389/frvir.2021.647997>

Interface as a tool

Collecting

Sensors generate and transmit their output in the form of unstructured signals / curated datasets and API

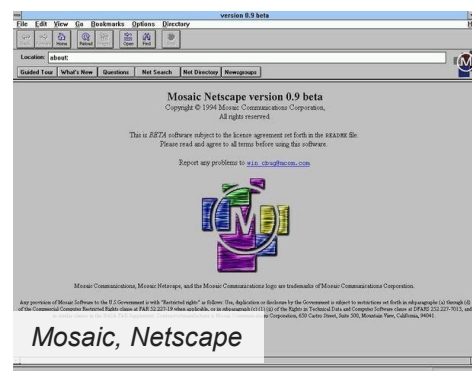
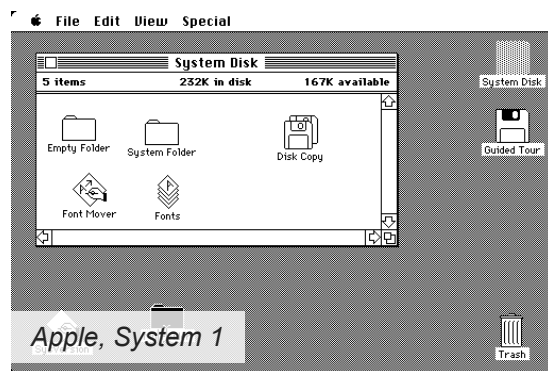
Processing

Data must be cleaned, selected, parsed, calibrated, synchronised, selected and transformed into intelligible information

Communicating

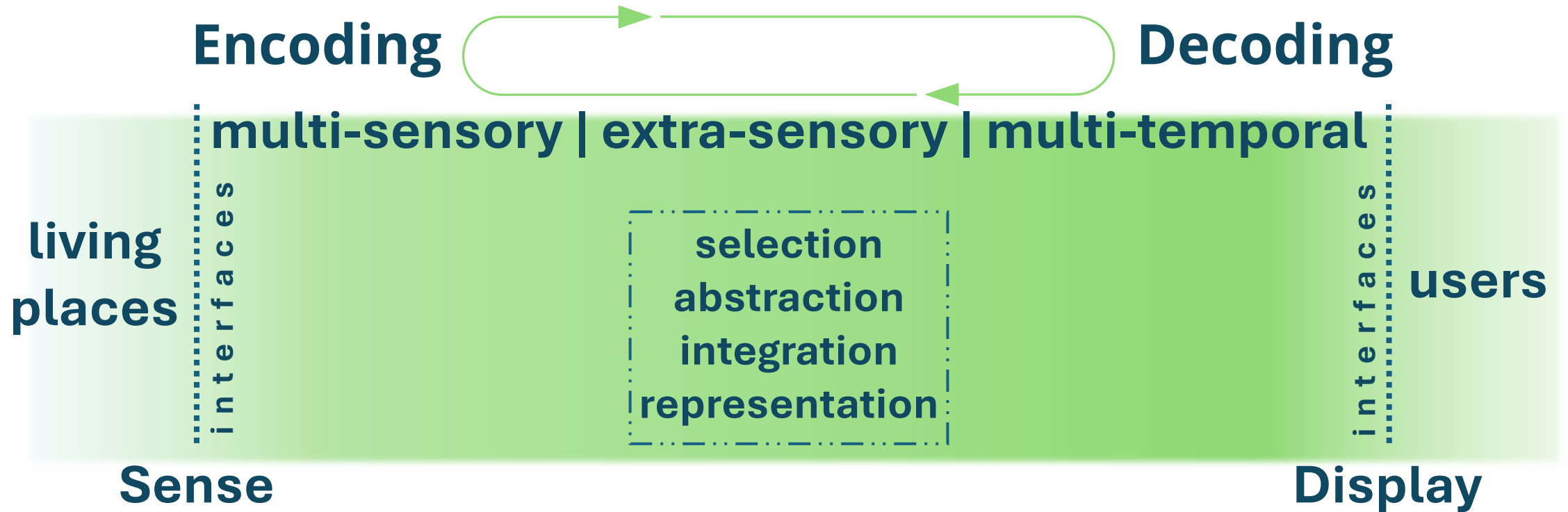
Information is transmitted to a system or human observer, and it is actionable to deliver value.

[Tech] An interface enables communication and interaction between systems, components, or users by providing a structured layer that translates actions, inputs, or commands into understandable operations.



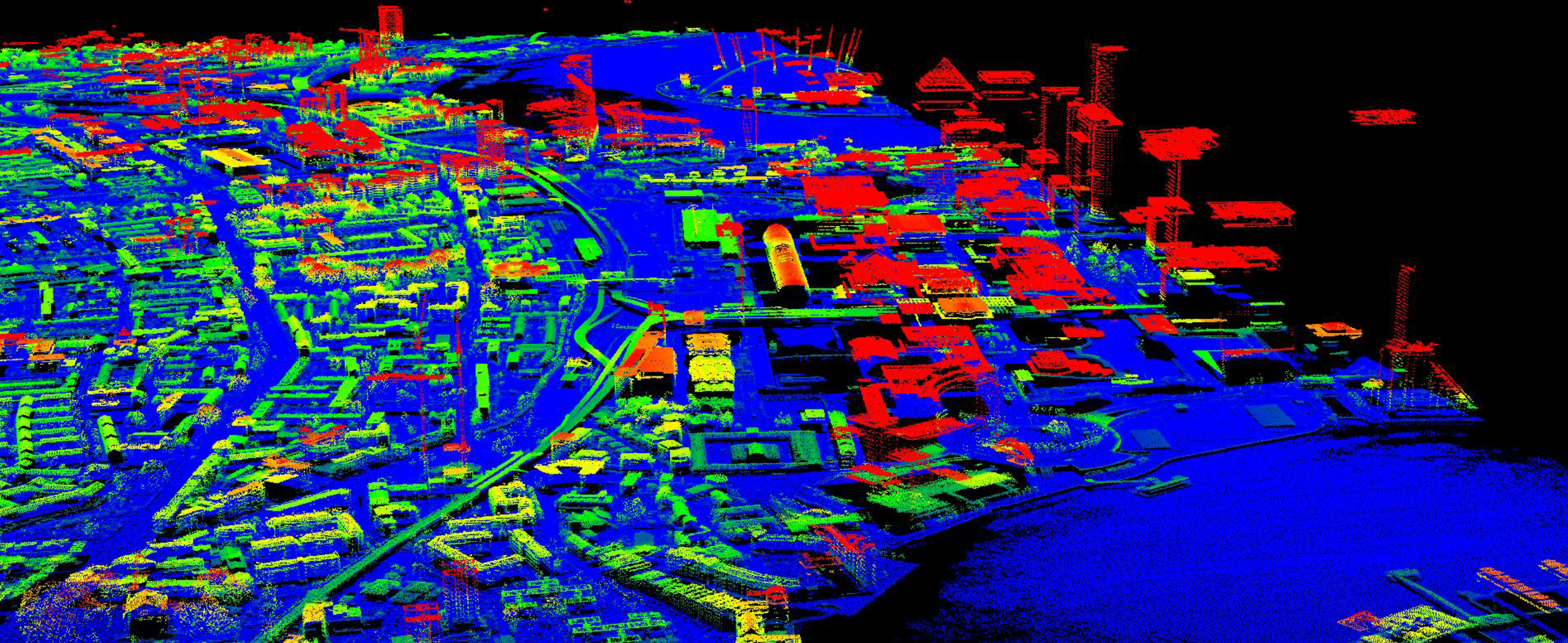
Interface as an effect

“The interface is this state of **‘being on the boundary’**. It is that moment where one significant material is understood as distinct from another significant material. In other words, an interface is not a thing, **an interface is always an effect. It is always a process or a translation.**” (Galloway, 2008:33)



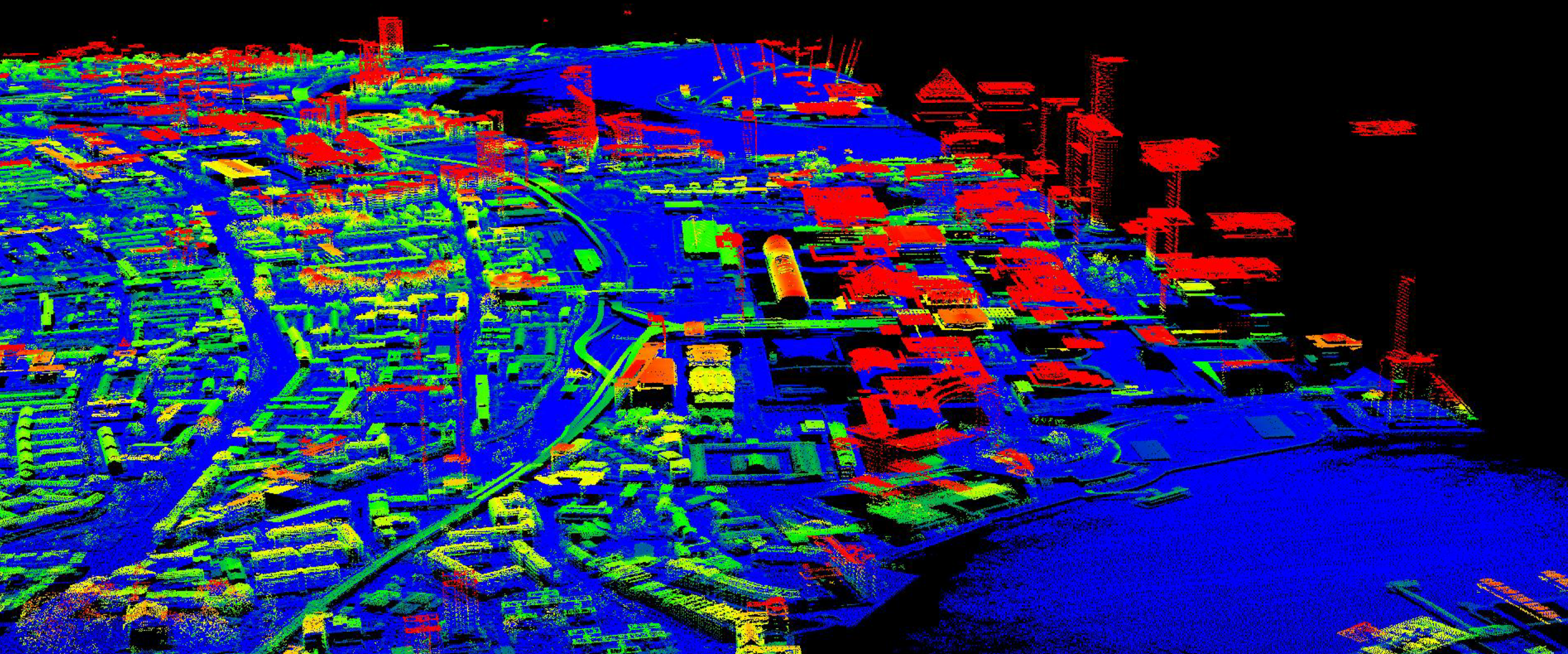
E/D continuum by Signorelli V. 2021. CC BY 4.0

Data Visualisation Approaches



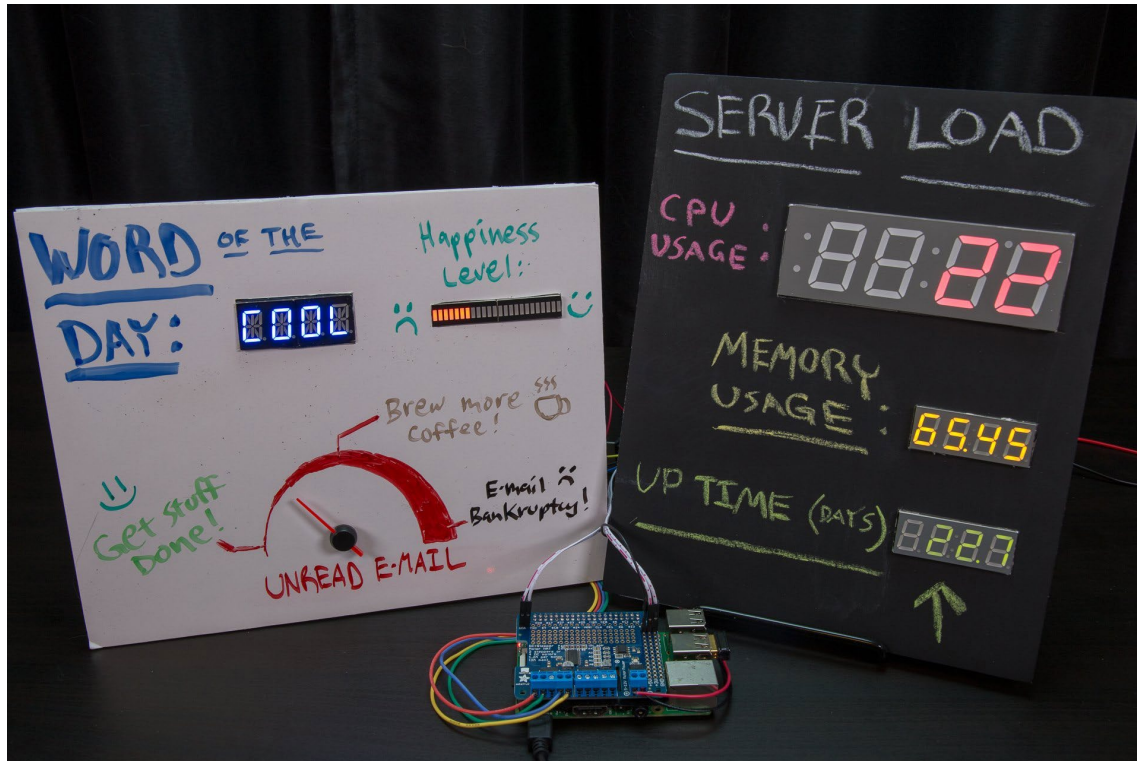
- **Dashboards** | collections of charts and visual elements organised to summarise the performance of a system.
- **Digital Twins (-ish)** | IoT data embedded in 3D models / abstract schematic diagrams of systems.
- **Situated Visualisation** | data visualised on the object or location of the sensor itself.

Dashboard



Dashboard

a system used to display various types of visual data in one place



Tony DiCola, CC BY-SA 3.0, via [Adafruit Learning System](#)

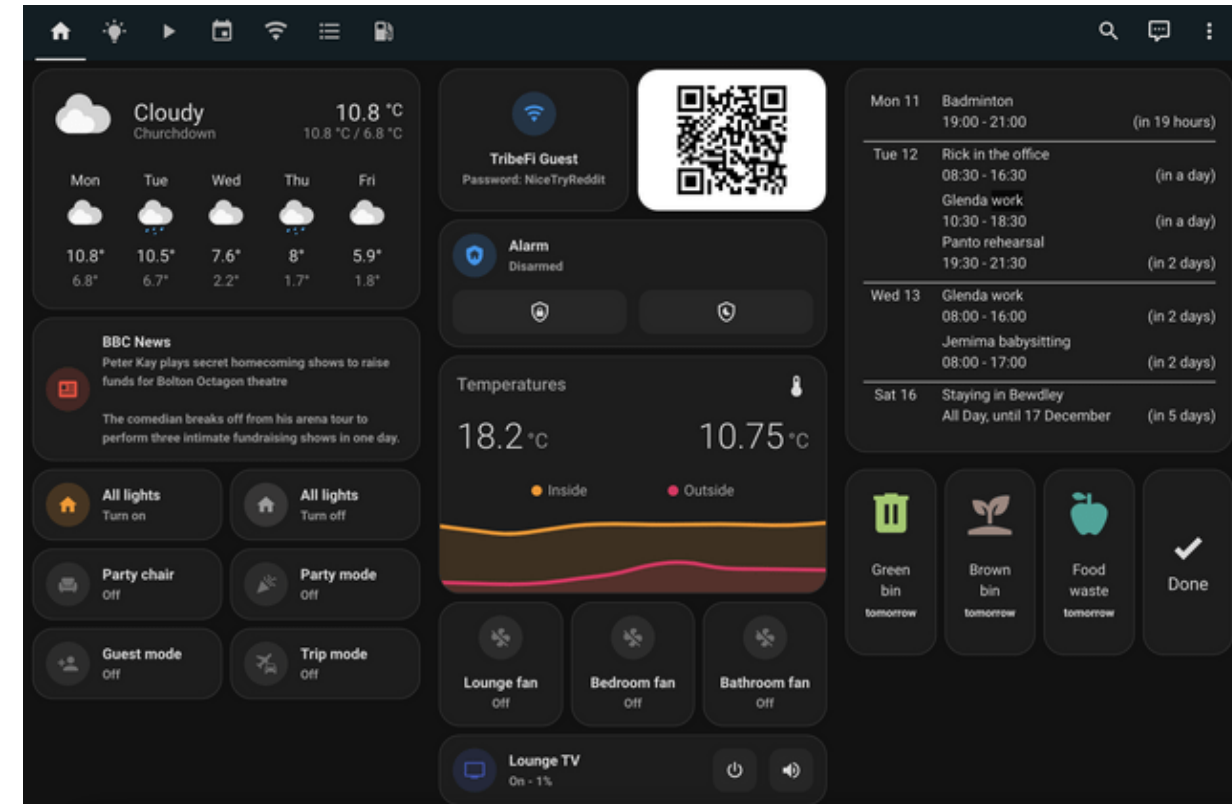


Duch.seb, CC BY-SA 3.0, via [Wikimedia Commons](#)

Dashboard

It needs to address a specific questions

- **Why** – the purpose of the dashboard
- **Who** – the target user of the dashboard
- **What** – the datasets and post-processing data
- **How** – the visualisation forms to be used



Source: HomeAssistant Dashboard, by Tribal2 on [Reddit](#)

When and Where can be also used but there are many other more detailed questions that can be asked as well

Applications range from raw data visualisation to advanced analytics, providing insights into performance, structure, patterns, and trends

- Weather monitoring

Home Automation

Fitness tracking

BIM

City Dashboard

Customer metrics
- Financial information

Marketing information

Web analytics

Energy monitoring

Human resources data

Logistics information
- Risk management

Environmental

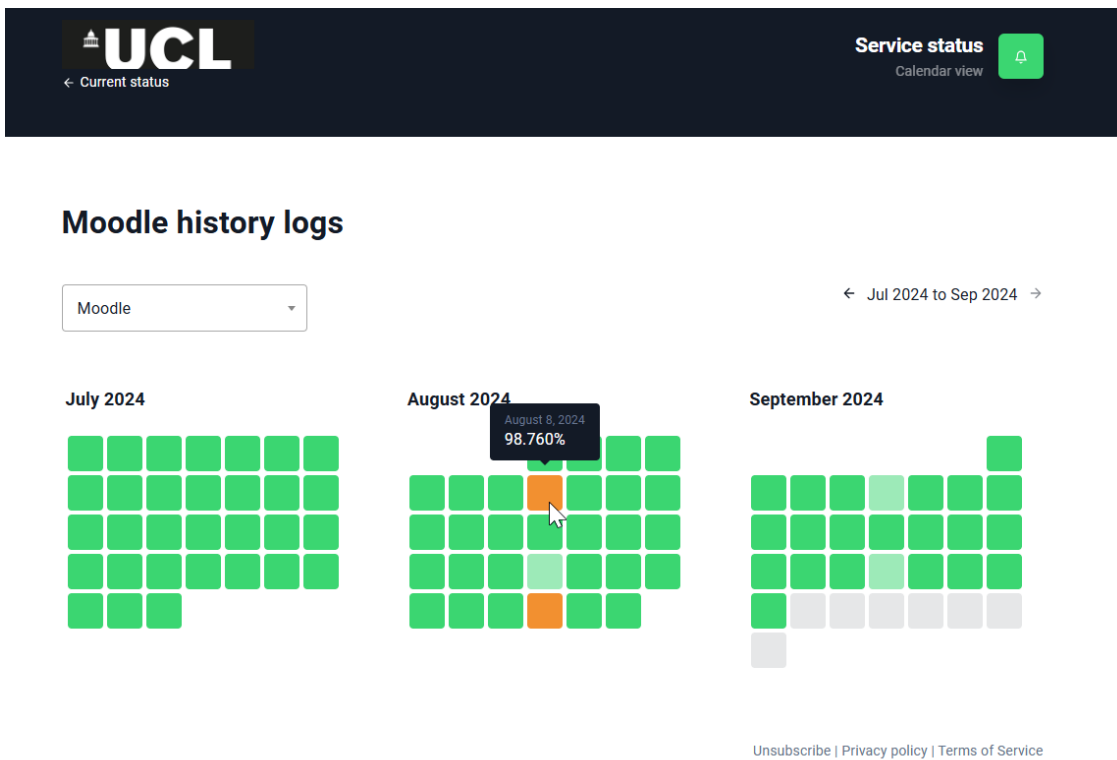
Public transport and more...

Data from multiple sources, different timescales, and various locations drive these insights

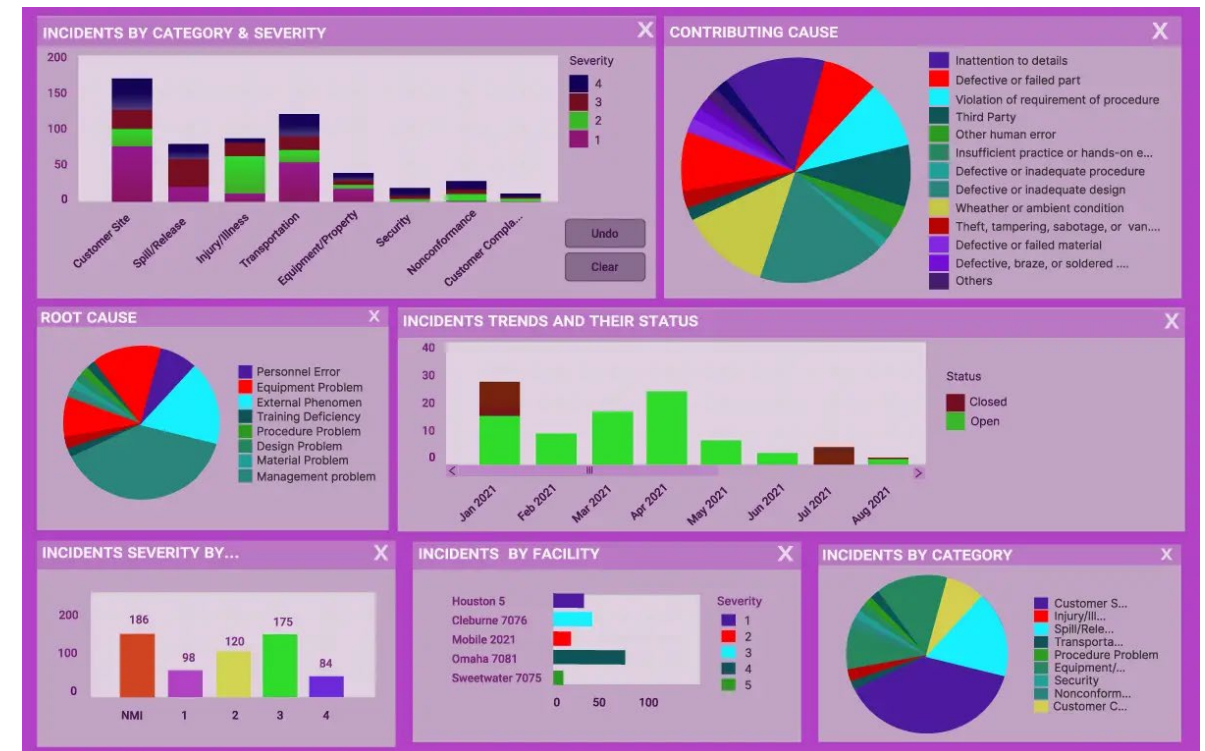
District	Part closure	+
DLR	Part closure	+
Elizabeth line	Part closure Reduced service	+
London Overground	Part closure	+
Metropolitan	Planned closure	+
Piccadilly	Part closure	+
Waterloo & City	Planned closure	+
Bakerloo	Good service	
Central	Good service	
Circle	Good service	
Hammersmith & City	Good service	

Dashboards do not need to be complex!

Effective dashboards are simple, clear, and easy to understand



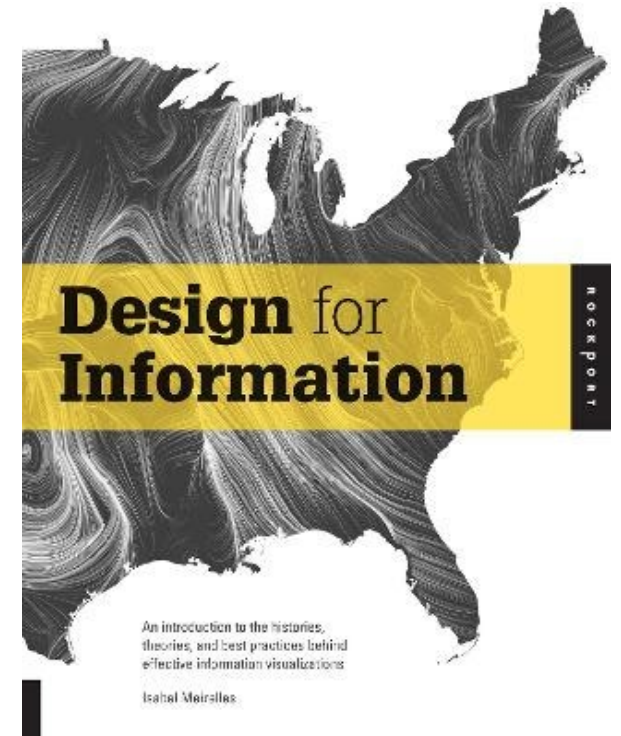
UCL ISD Systems



Design principles

Some are common to other UI and visualisations, others are more specifically focused on dashboards

- **Easy to Read:** content should be quickly readable (effective organization)
- **Concise Information:** limit text, ensure clarity and informative content
- **Aesthetic:** balance aesthetics with efficiency
- **Colours and Contrast:** enhance readability and accessibility
- **Interactivity:** include easy-to-understand controls if needed
- **Detail Balance:** provide balanced insights in the frontend view
- **Visualisation Objects:** use appropriate objects to represent data



Isabel Meirelles (2013) *Design for Information*. Rockport Publishers.

Most common visualisation objects (but not limited to)

- Gauges
- Charts
- Tables
- Text
- Maps



How to choose the right chart?

The Data Visualisation Catalogue

About • Blog • Shop • Resources

What do you want to show?

Here you can find a list of charts categorised by their data visualization functions or by what you want a chart to communicate to an audience. While the allocation of each chart into specific functions isn't a perfect system, it still works as a useful guide for selecting chart based on your analysis or communication needs.



Comparisons



Proportions



Relationships



Hierarchy



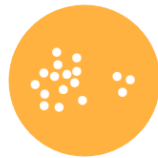
Concepts



Location



Part-to-a-whole



Distribution



How things work



Processes & methods



Movement or flow



Patterns

The Data Visualisation Catalogue

About • Blog • Shop • Resources

Comparisons

Visualisation methods that help show the differences or similarities between values.

With an axis



Bar Chart



Box & Whisker Plot



Bubble Chart



Bullet Graph



Histogram



Line Graph



Marimekko Chart



Multi-set Bar Chart



Nightingale Rose Chart



Parallel Coordinates Plot



Population Pyramid



Radar Chart



Radial Bar Chart



Radial Column Chart



Span Chart



Stacked Area Graph



Stacked Bar Graph

Without an axis



Chord Diagram



Choropleth Map



Donut Chart



Dot Matrix Chart

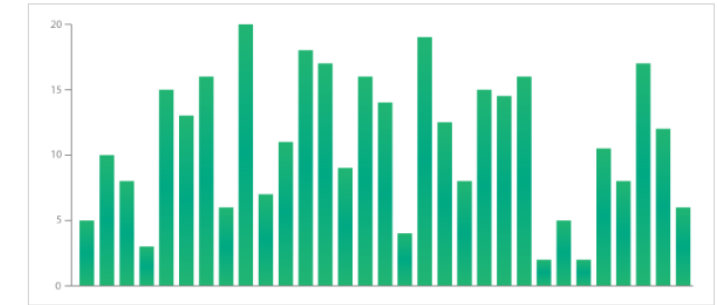


Heatmap



Parallel Sets

Bar Chart



Description

As known as *Bar Graph* or *Column Graph*.

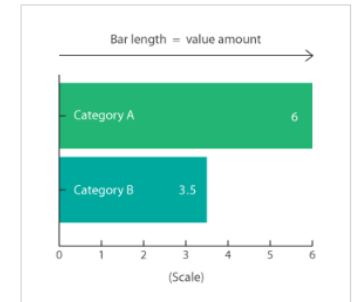
A Bar Chart uses either horizontal or vertical bars (column chart) to show discrete, numerical comparisons across categories. One axis of the chart shows the specific categories being compared and the other axis represents a discrete value scale.

Bar Charts are distinguished from Histograms, as they do not display continuous developments over an interval. Instead, Bar Chart's discrete data is categorical and therefore answers the question of "how many?" in each category.

Functions

Comparisons Patterns

Anatomy



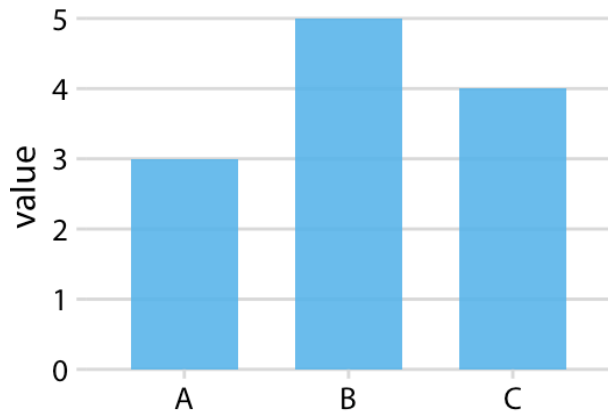
Similar Charts



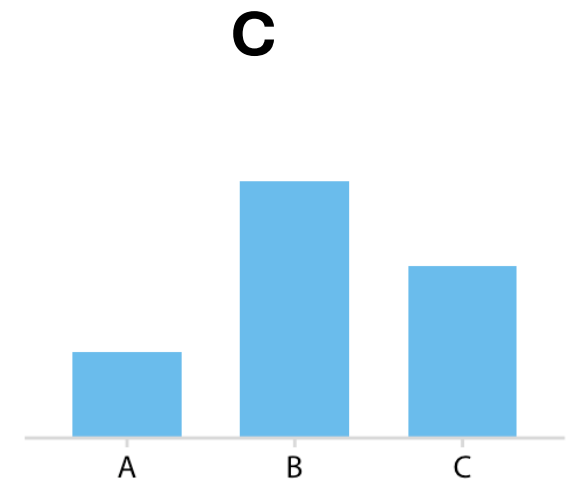
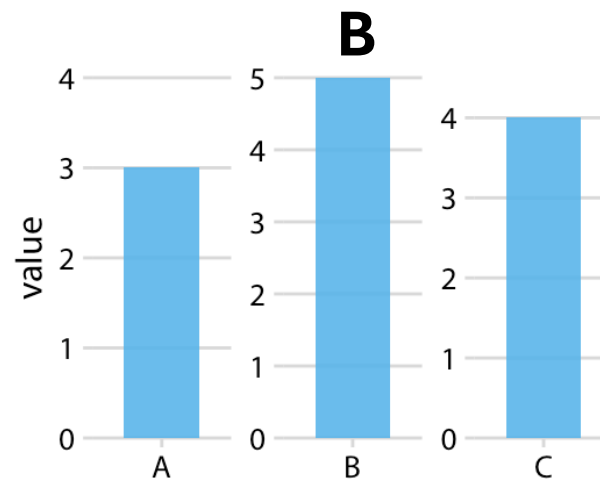
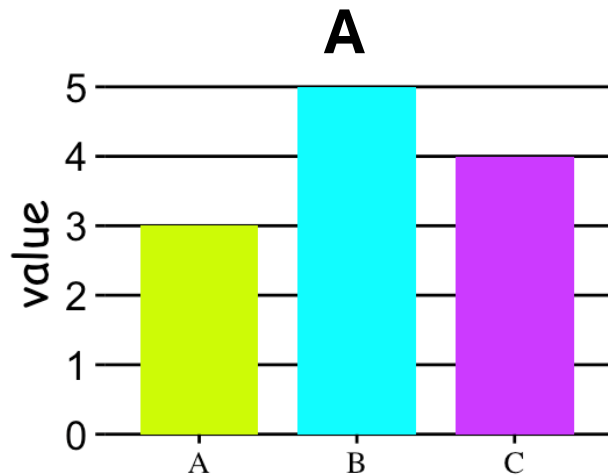
Tools to Generate Visualisation

Code-based:
[AnyChart \(JS\)](#)

Dashboard

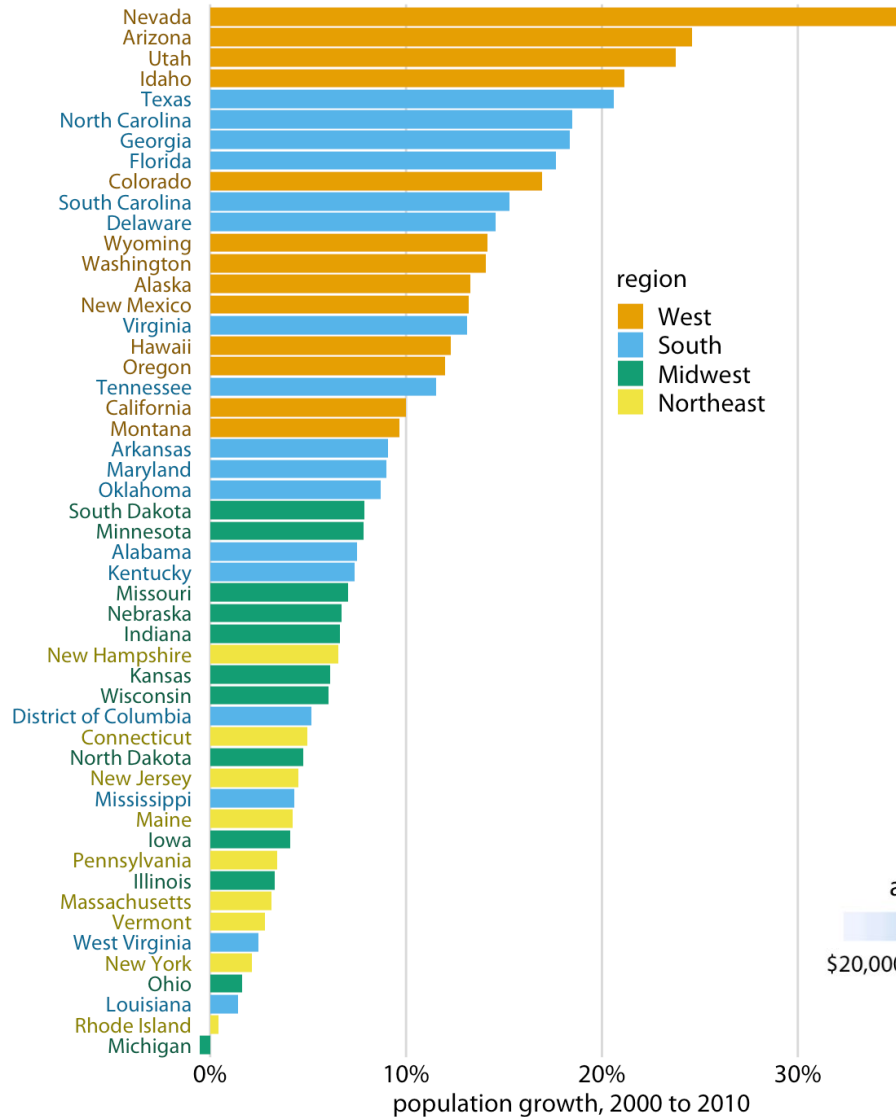


- **Title and sources**
- **Axes and Values** (scale and formatting)
- **Legend** (not always needed)
- **Colours** (not just aesthetic)



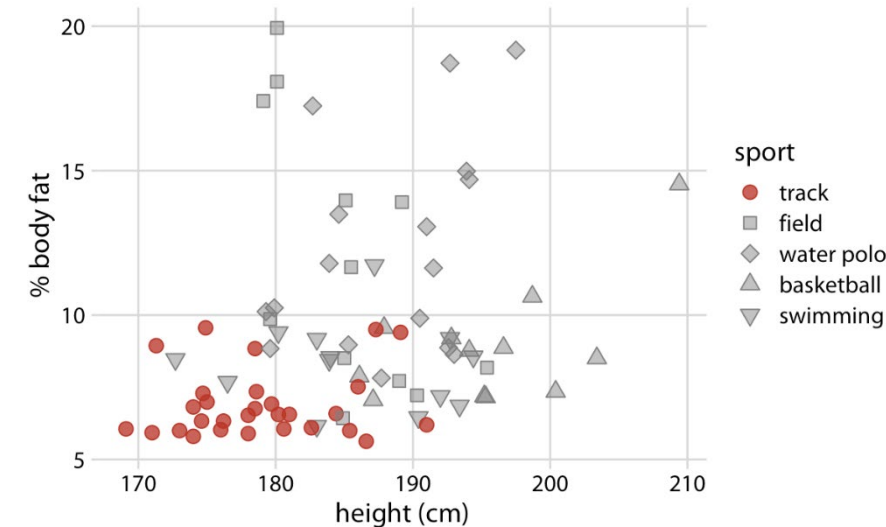
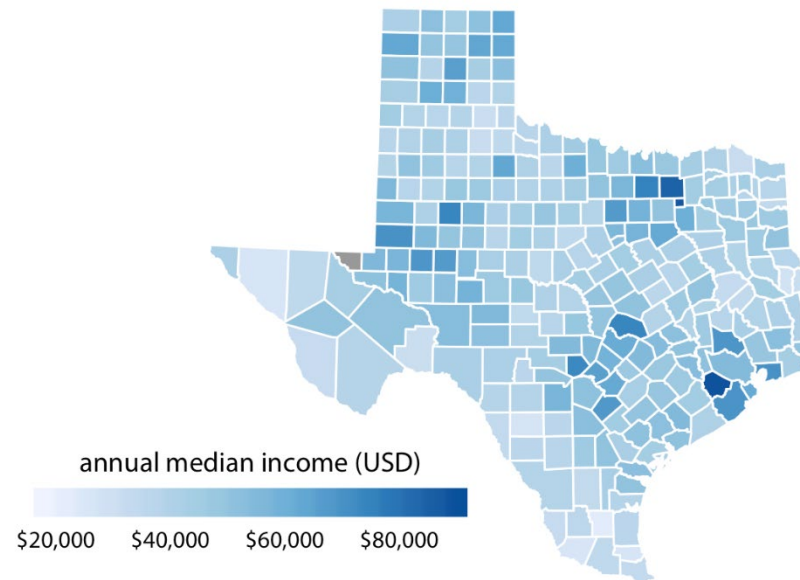
Wilke, C. O. (n.d.). *Fundamentals of Data Visualization* <https://clauswilke.com/dataviz/>

Dashboard



Colours to...

- Distinguish
- Represent
- Highlight



Wilke, C. O. (n.d.). *Fundamentals of Data Visualization* <https://clauswilke.com/dataviz/>

Apache ECharts

An Open Source JavaScript Visualization Library



<https://echarts.apache.org>

Wilke, C. O. (n.d.). *Fundamentals of Data Visualization* <https://clauswilke.com/dataviz/>



Bloomberg Terminal. Travis Wise, CC BY 2.0

From passive to interactive interfaces

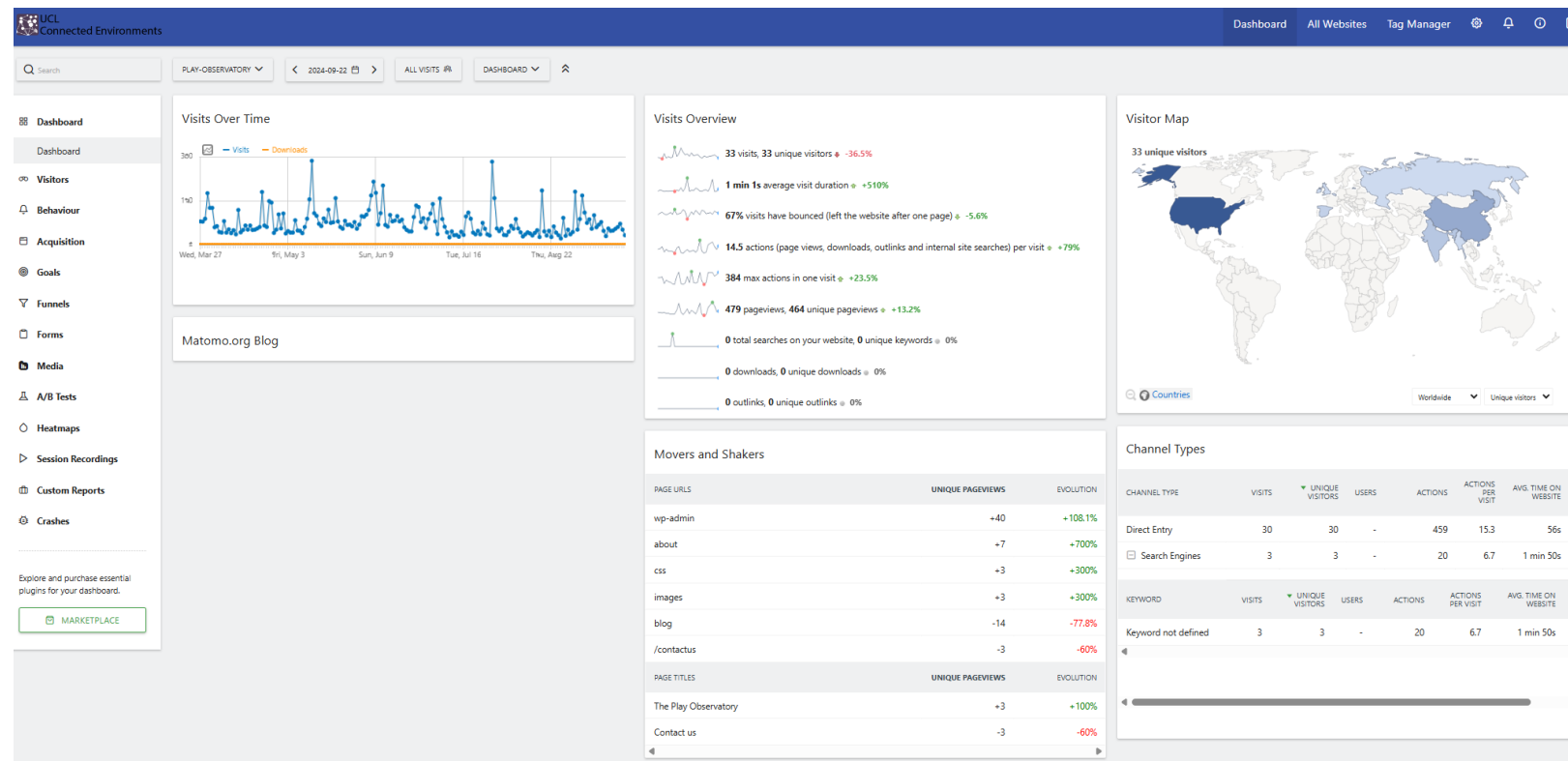
- 80s - financial monitor - Bloomberg Terminal
- 90s - beyond financial - CompStat NYPD; Baltimore CitiStat

Data science field to expand with data-driven approaches

Beyond accumulation of data, but data to drive decisions and strategies

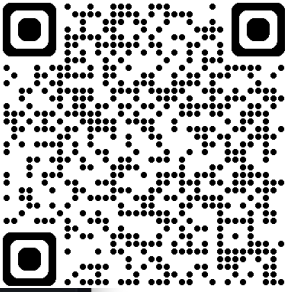
Shannon Mattern, “Mission Control: A History of the Urban Dashboard,” *Places Journal*, March 2015 <https://doi.org/10.22269/150309>

Web analytics

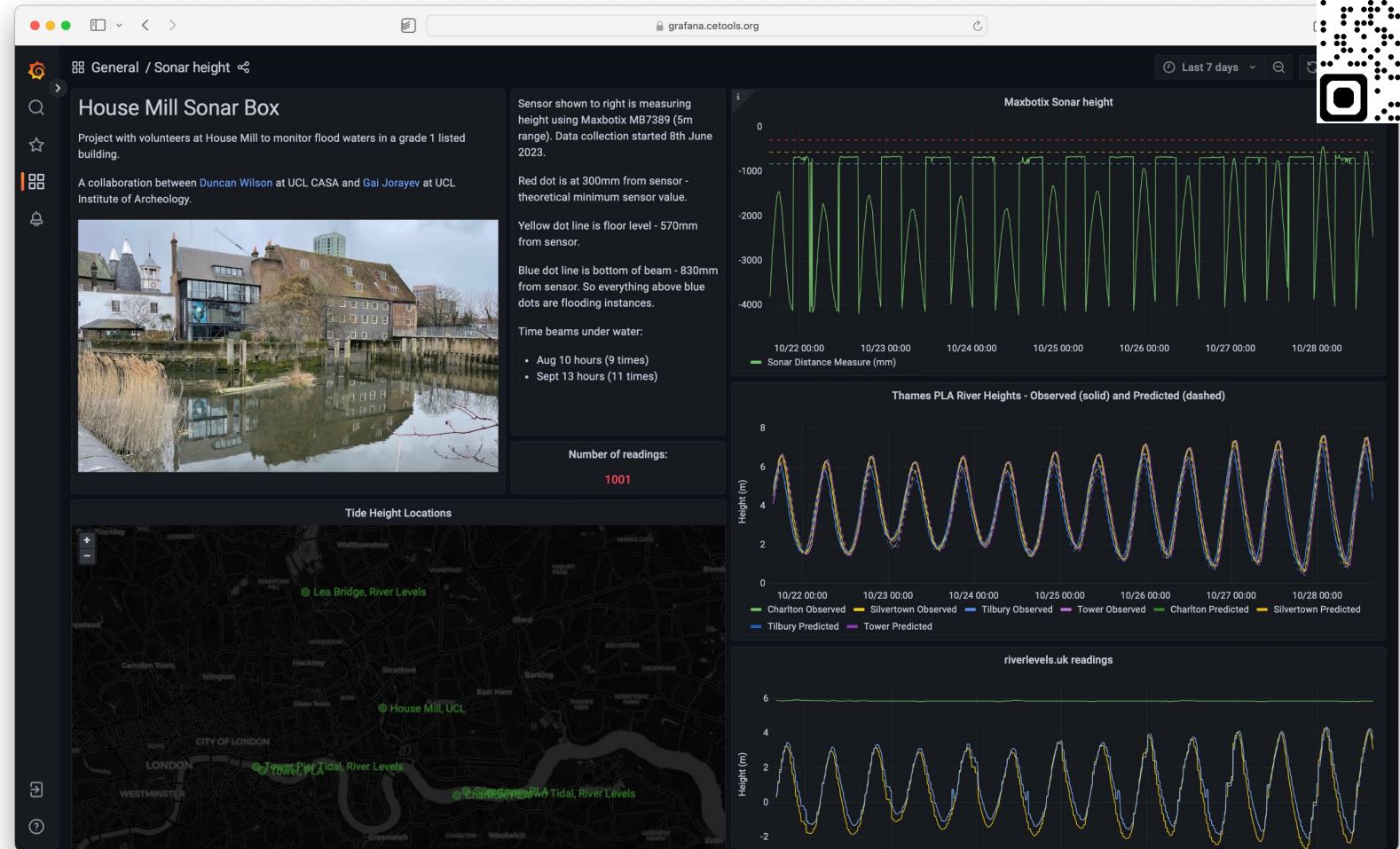


- Measuring impact
- Understand user behaviour
- Localisation
- Effective media
- Technical infrastructure

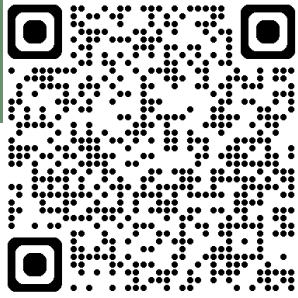
Web analytics of the play-observatory.com blog via Matomo



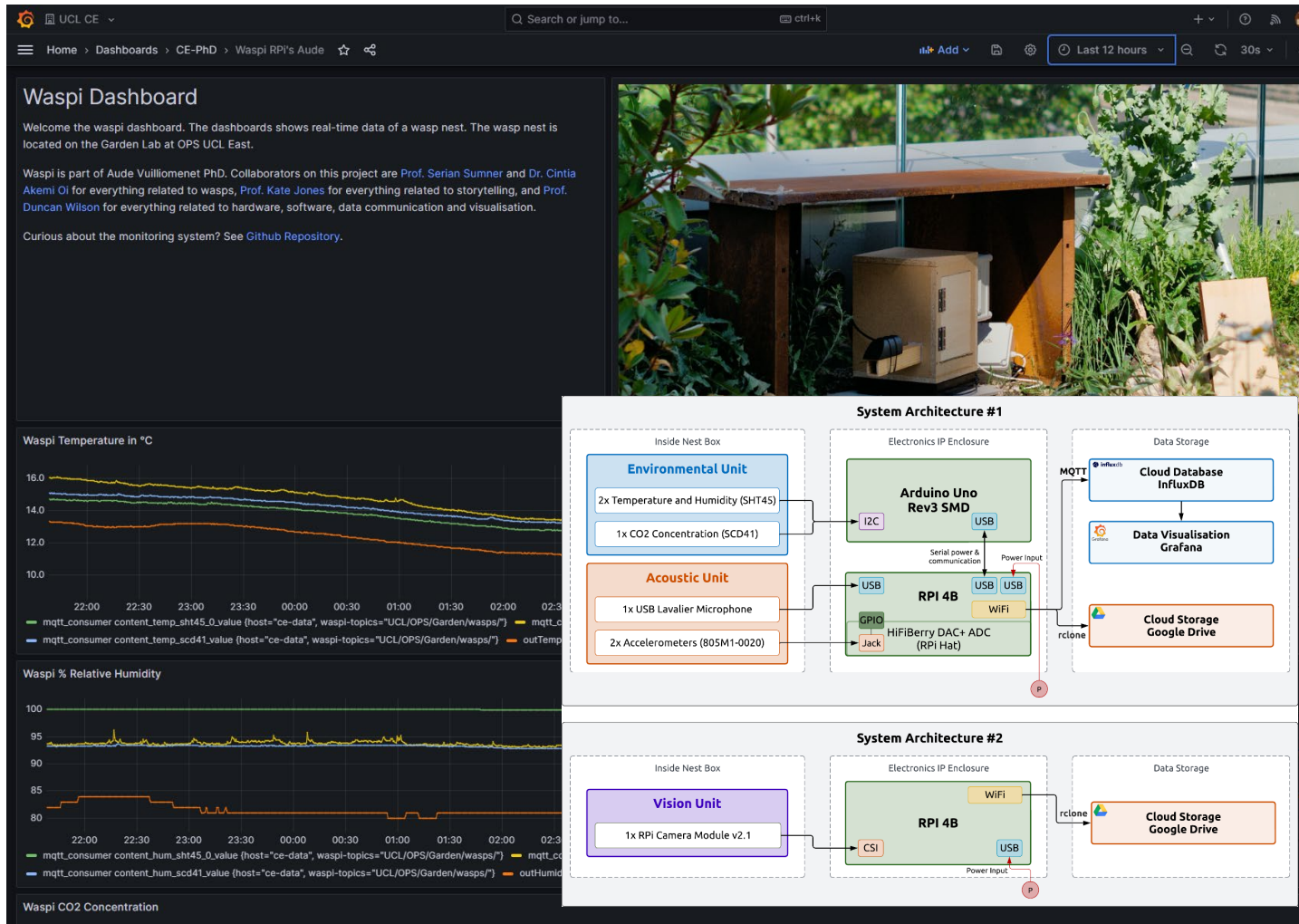
RT Data Visualisation



Wilson, D. (2023) 'House Mill flood monitoring', [iot.io](https://www.iot.io/blog/2023/10/14/housemill-flood.html), 14 October. Available at: <https://www.iot.io/blog/2023/10/14/housemill-flood.html> (Accessed: 20 July 2024)



RT Data Visualisation



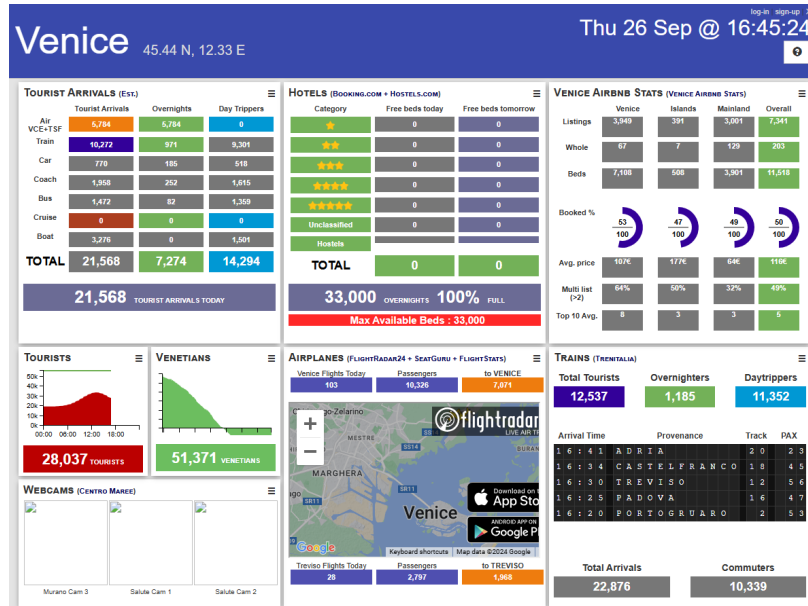
Waspi - IoT Monitoring System from Aude Vuilliomenet

Real-time wasp nest monitoring to understand ecology and behaviour of wasps

- Temperature and humidity sensors reveal how wasps regulate their home
- weight scale tracks nest growth
- sound sensors capture their buzzing and scratching
- camera records their comings and goings

Dashboard

To visual analytics to display and communicate information about the performance, structure, and patterns and trends of cities (e.g. gauges, meters, arrows, charts, graphs, maps)



Venice Dashboard (2013) <https://dashboard.cityknowledge.net/#/venice>



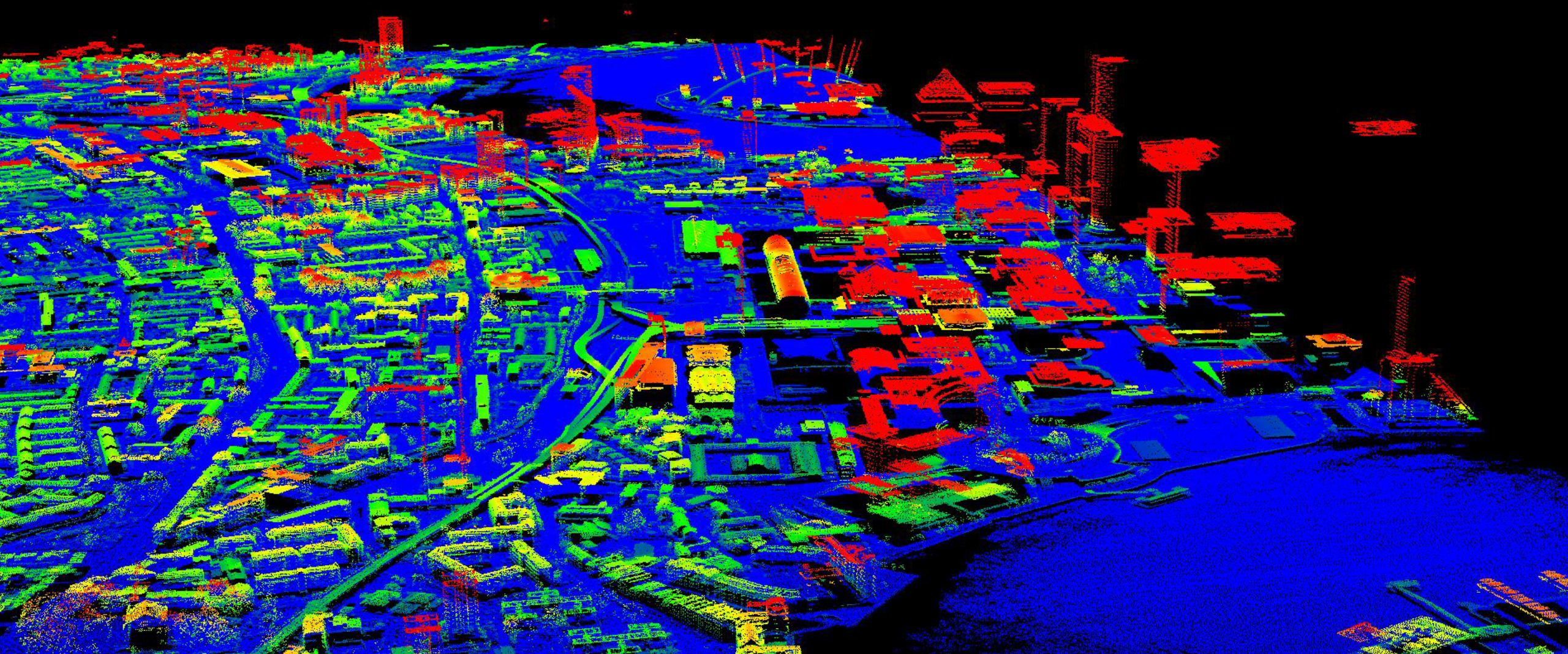
CityDashboard (2012) <https://citydashboard.org/london>

Dashboard are not neutral, they are framed socially, politically, ethically, philosophically in terms of their form, selection of data, modes of display and analysis, and deployment.

Kitchin, R., Maalsen, S., & McArdle, G. (2016). The praxis and politics of building urban dashboards. *Geoforum*, 77, 93–101. <https://doi.org/10.1016/j.geoforum.2016.10.006>

Young, G. W., & Kitchin, R. (2020). Creating design guidelines for building city dashboards from a user's perspectives. *International Journal of Human-Computer Studies*, 140, 102429. <https://doi.org/10.1016/j.ijhcs.2020.102429>

Digital Twin and BMS



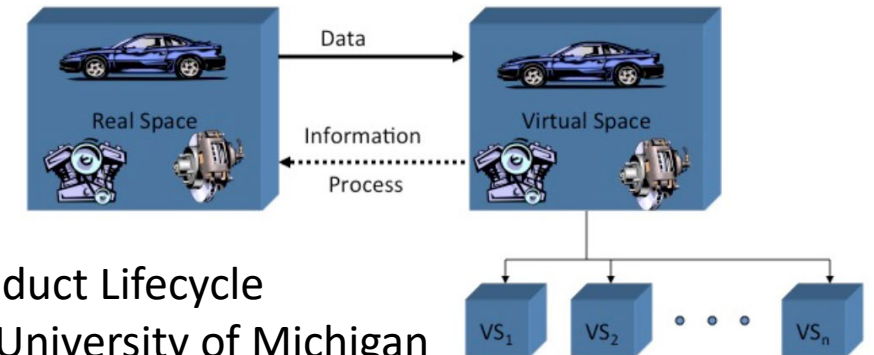
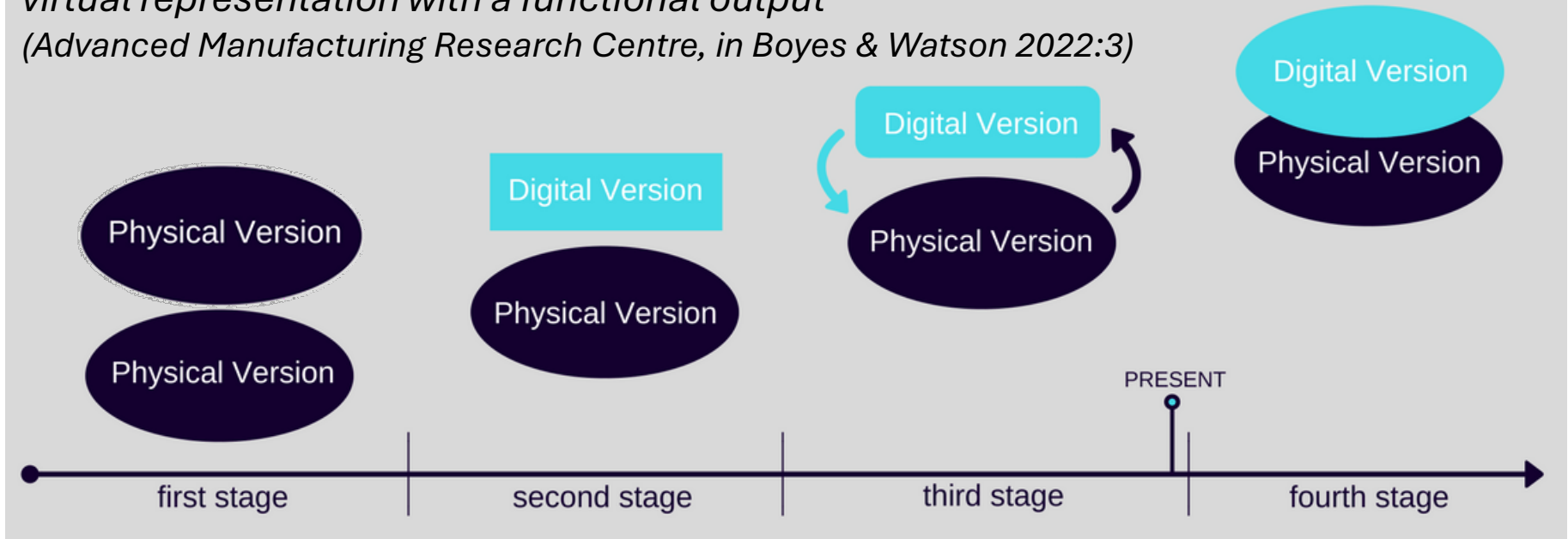
From MIRRORED SPACES to DIGITAL TWINS



Jim Lovell descends from the Command Module simulator at KSC in 1970

“A live digital coupling of the state of a physical asset or process to a virtual representation with a functional output”

(Advanced Manufacturing Research Centre, in Boyes & Watson 2022:3)



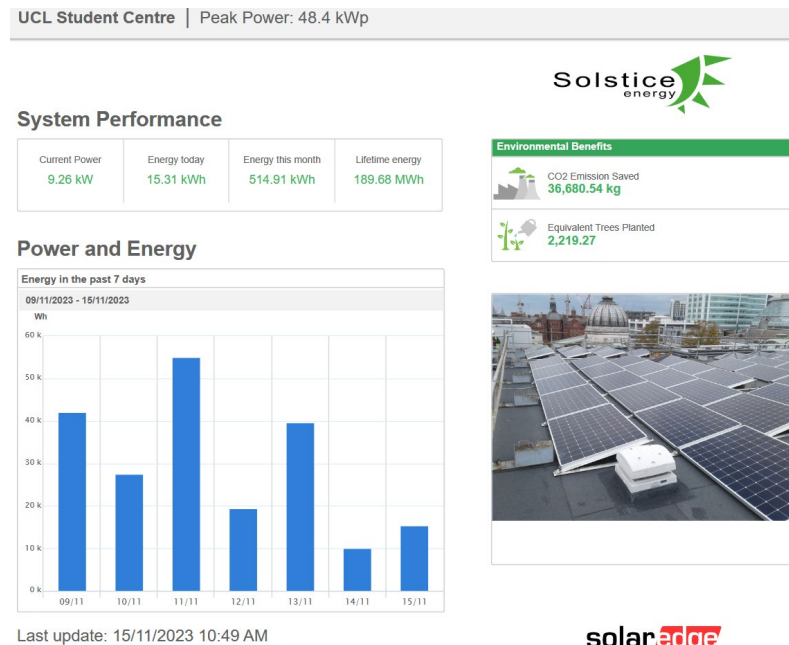
M. Grieves, 2002, Product Lifecycle Management (PLM), University of Michigan

Images: <https://www.hq.nasa.gov/alsj/a13/images13.html> (l),
<https://www.challenge.org/insights/digital-twin-history> (r)

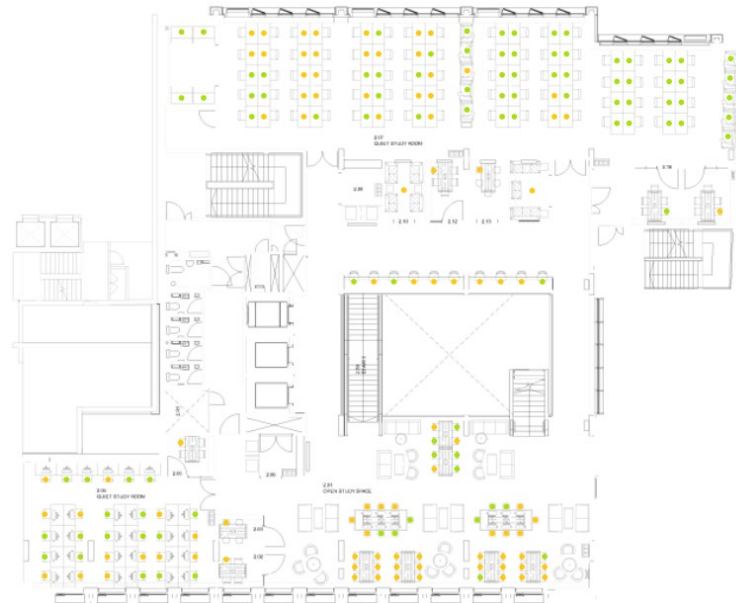
Digital Twin

Integrated systems but separate visualisations

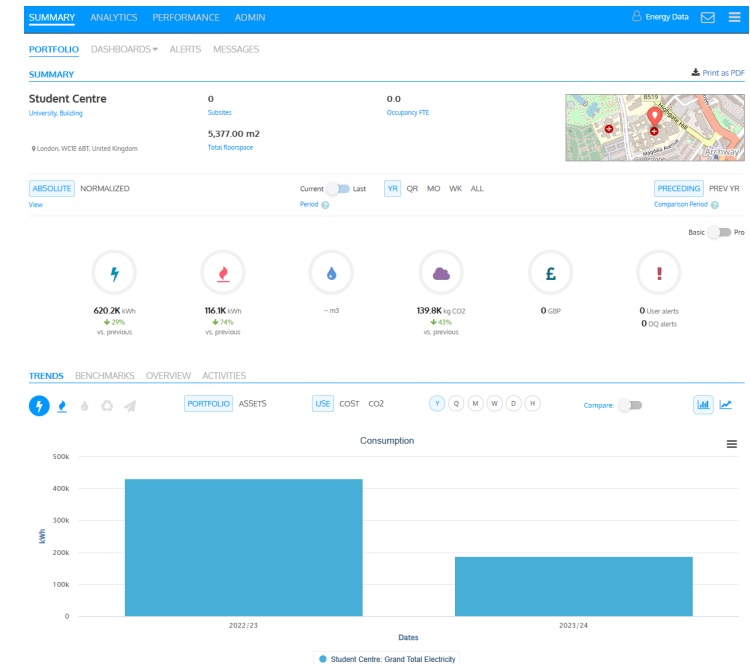
Same building, different system manager platforms, while data is often linked (e.g. energy production; energy consumption; occupancy) the overall understanding is missing



Solar Power Data



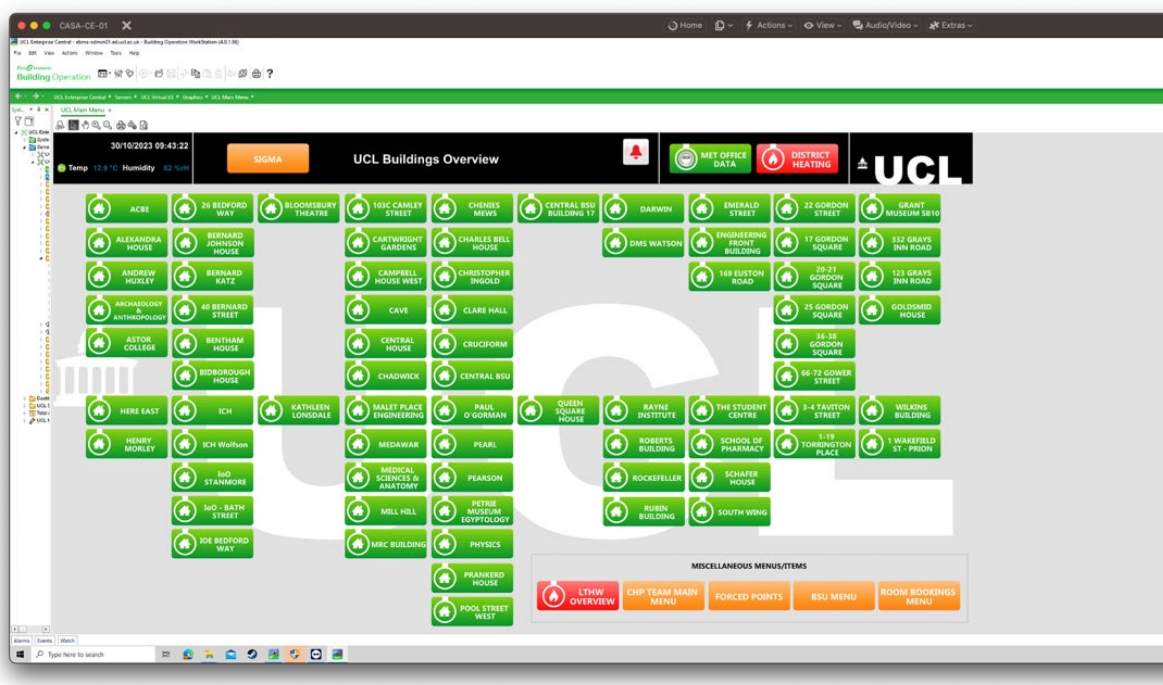
Occupancy data



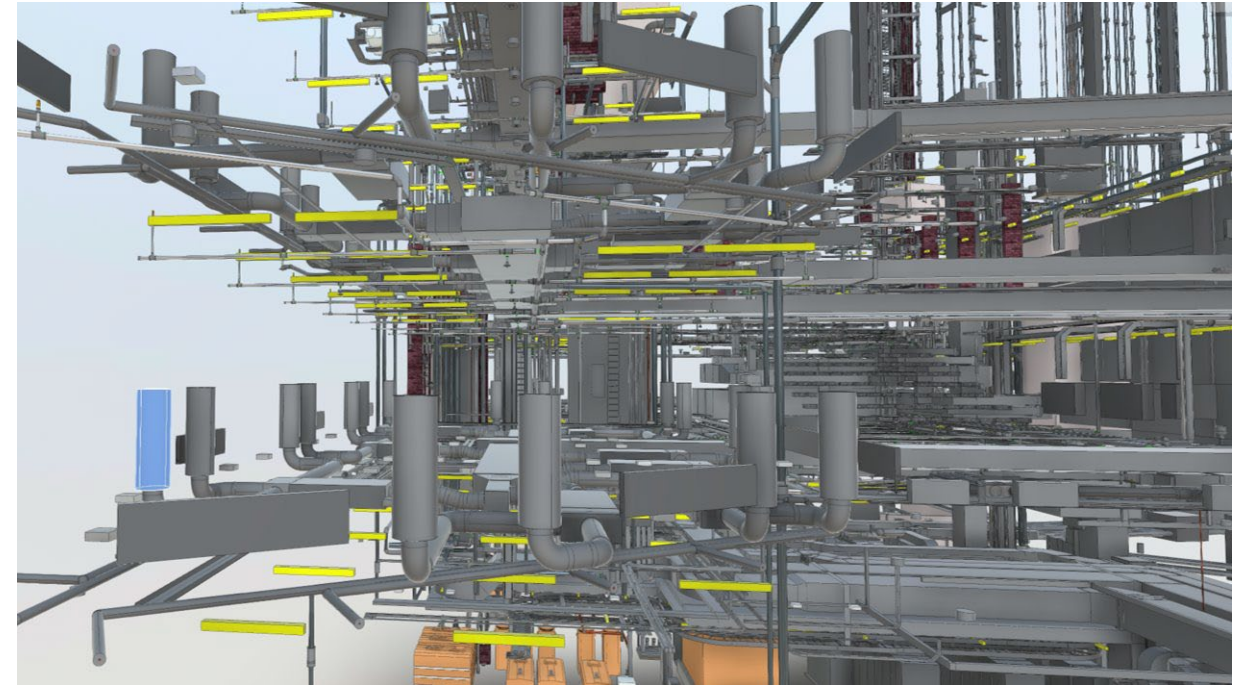
Energy Consumption

Abstract visualisations give overview of complex systems but are mostly usable by experts

BIM visualisations are clearer and detailed but heavy and hard to navigate



BMS Control panel



HVAC

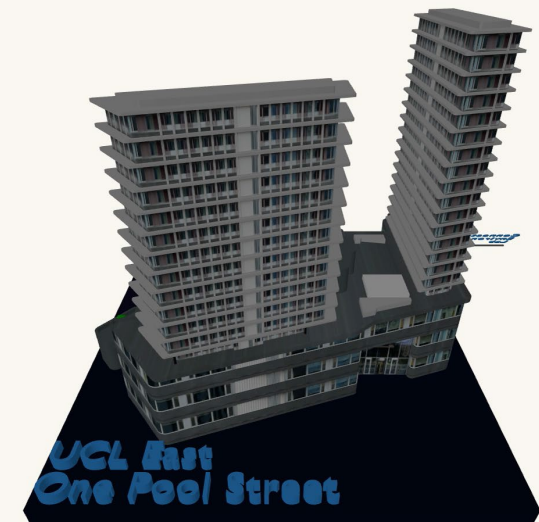
Digital Twin

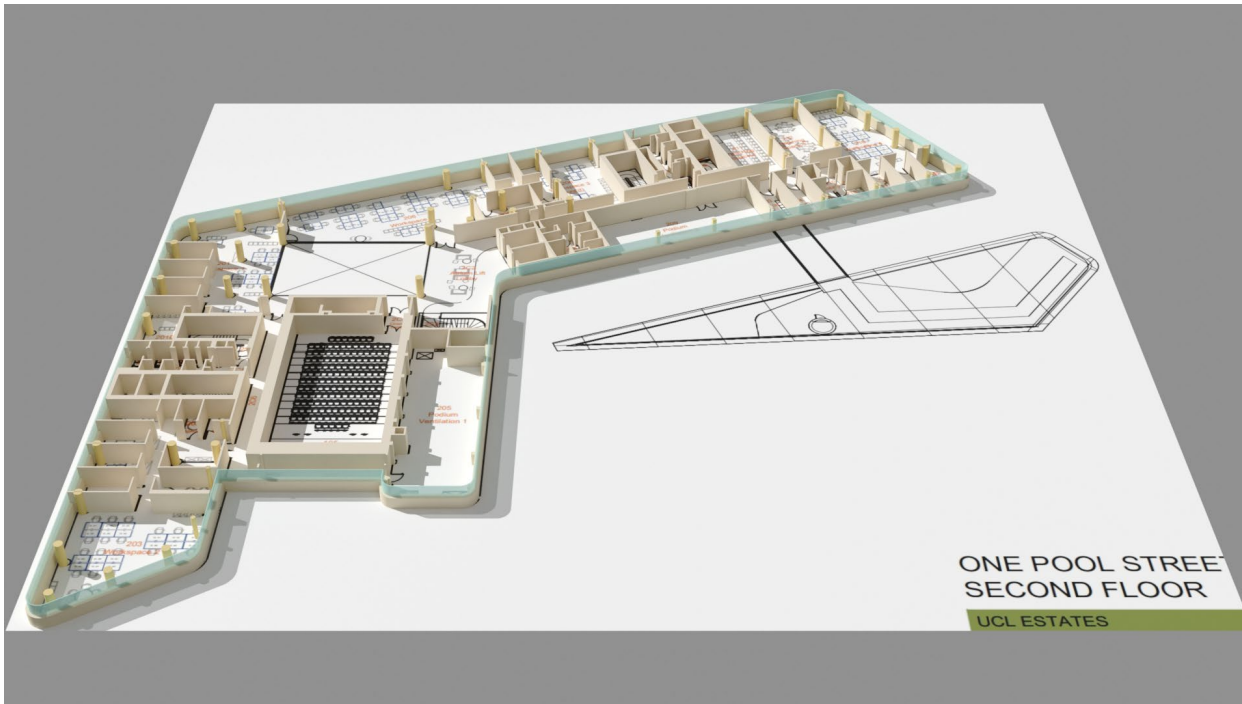


<https://gardenlab.cetools.org/>

3D model as a support for:

- Visualisation
- Planning
- Communication



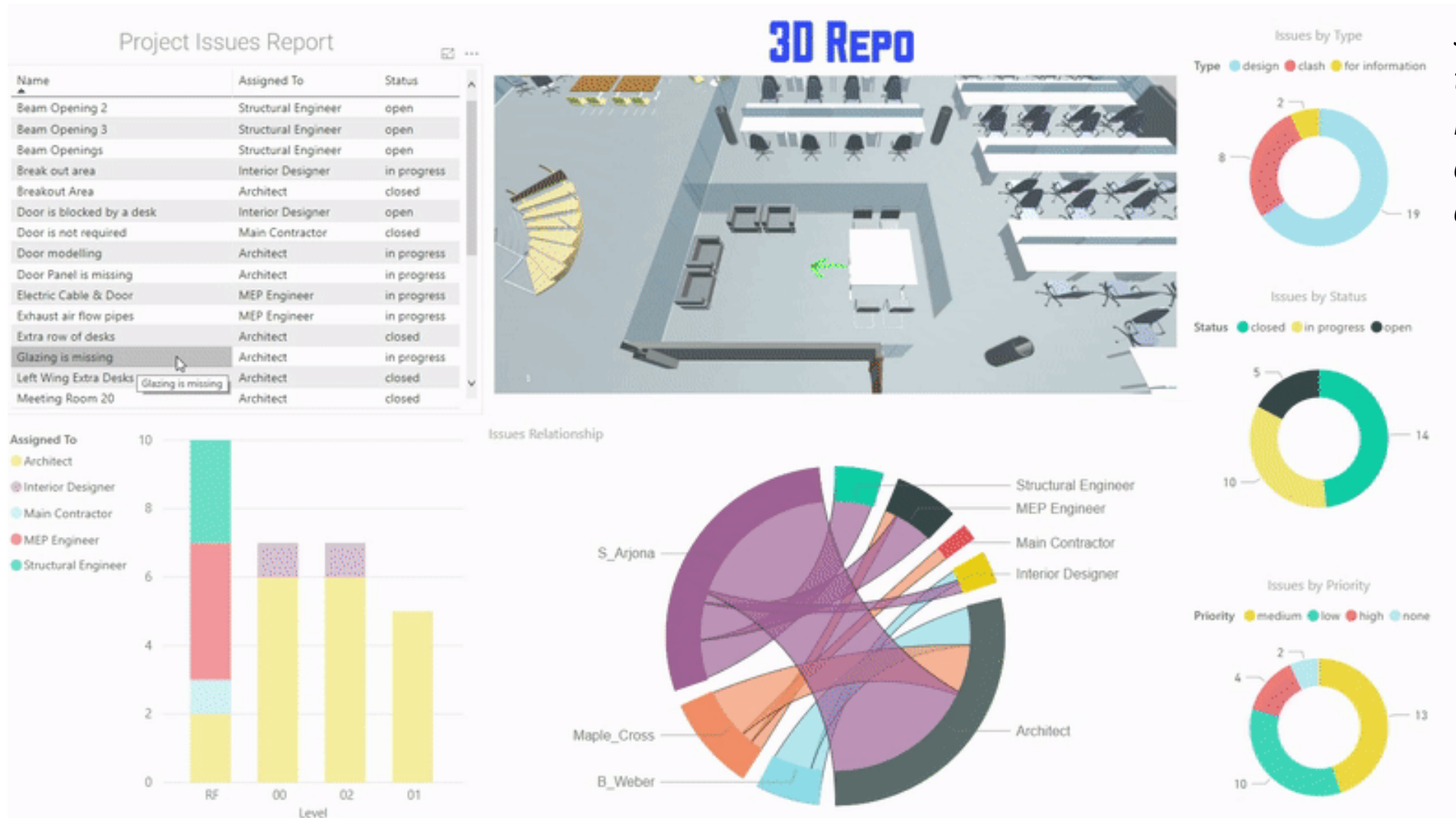


Work in progress

▼ UCL

► OPSEBOAS (6663 topics, 42113 messages)

Digital Twin



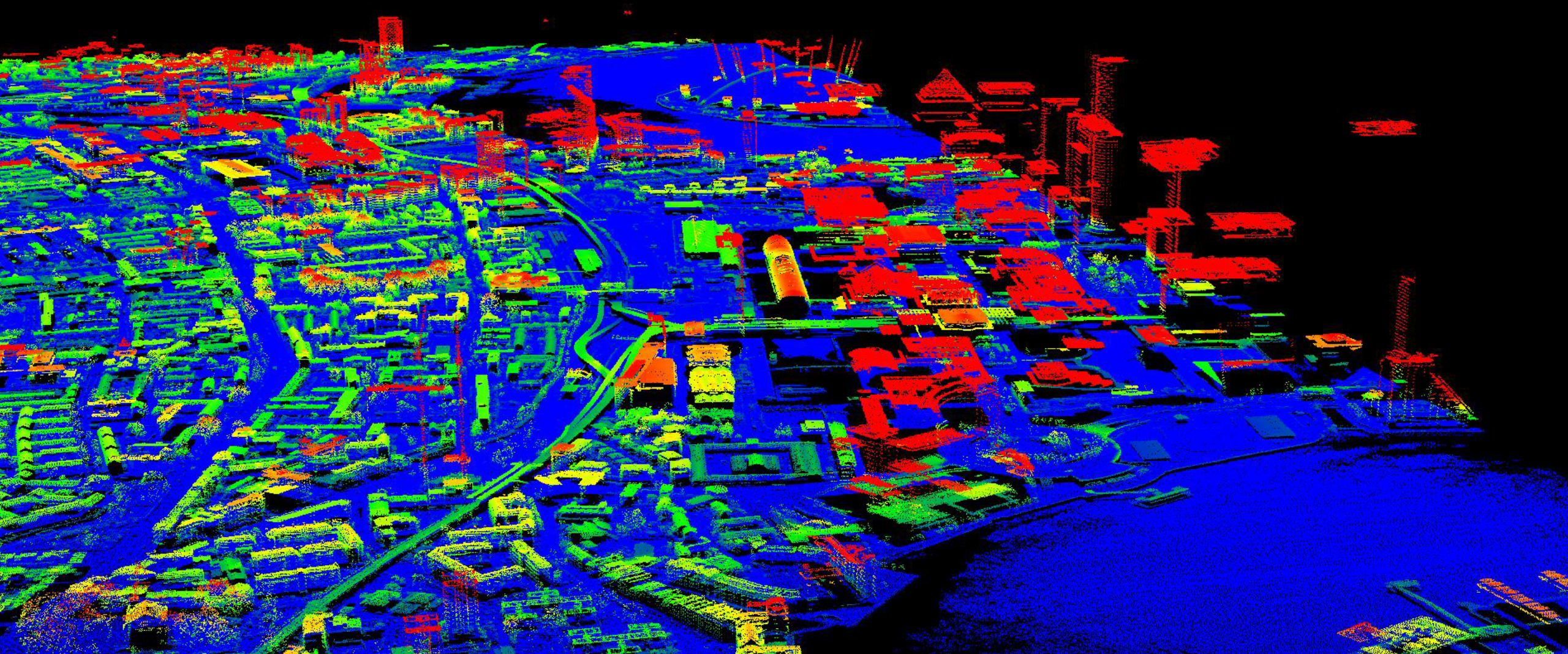
3DRepo is an online platform to help manage Building Information Modelling (BIM) data during design, delivery and operation.

<https://3drepo.com/>

Digital Twin



Situated Visualisation

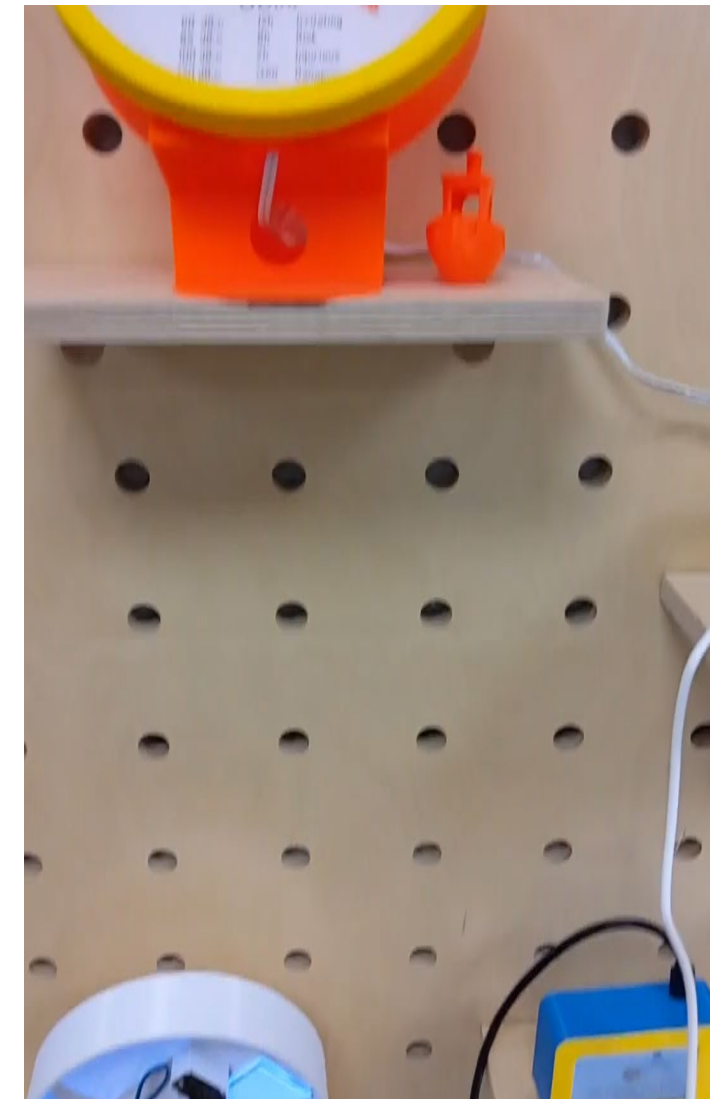


data displayed in context, within places or integrated in objects, that contribute to its interpretation and understanding

- **Contextual Relevance** *(it is about space and time)*
*Visualisations are **directly tied to the physical environment** and associating information with a specific place improves **memory retention** and recall.*
- **Enhanced Engagement** *(it is about consumption and participation)*
*Users **interact more with data in a familiar context** and improve collaboration; serves as **educational tools** for immersive learning.*
- **Improved Accessibility** *(it is about reach and diversity of users)*
***Diffuse visualisations** in the environment to make data **accessible to a broader audience** (though other aspects of accessibility need to be considered).*

Situated Visualisation

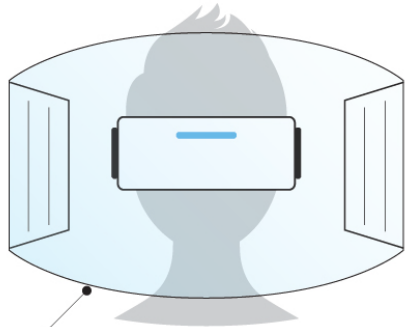
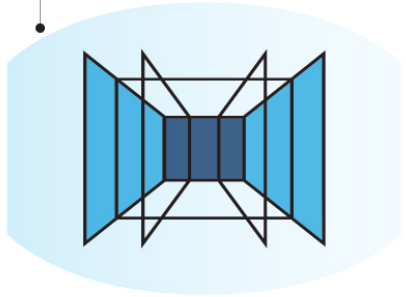
- **Situated spatially:**
 - Visualisations positioned in the physical environment, allowing data to be understood in relation to place.
- **Situated around the object:**
 - Augmenting the object with additional data that it cannot otherwise convey, enriching its meaning through layered information.



Situated Visualisation

VIRTUAL REALITY (VR)

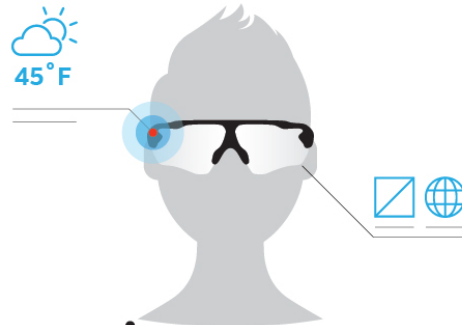
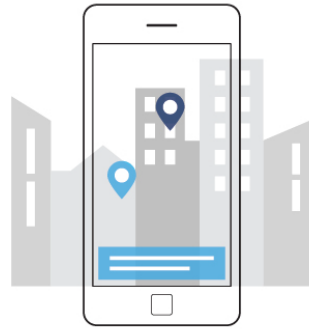
Completely digital environment



Fully enclosed, synthetic experience with no sense of the real world.

AUGMENTED REALITY (AR)

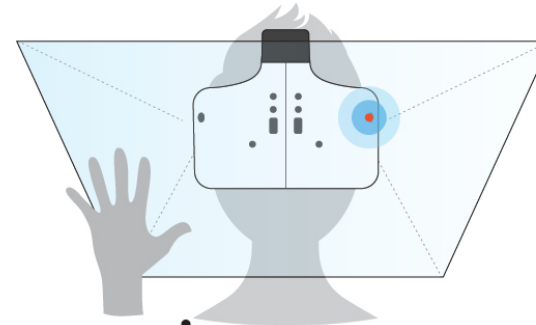
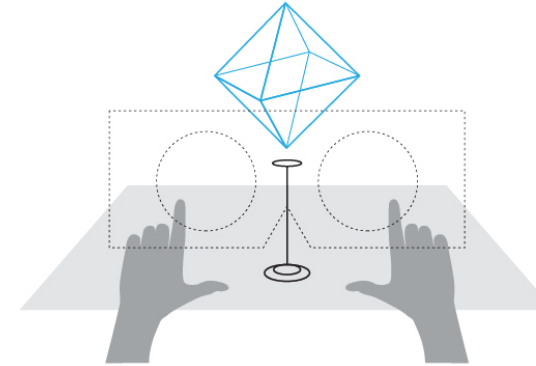
Real world with digital information overlay



Real world remains central to the experience, enhanced by virtual details.

MERGED REALITY (MR)

Real and the virtual are intertwined

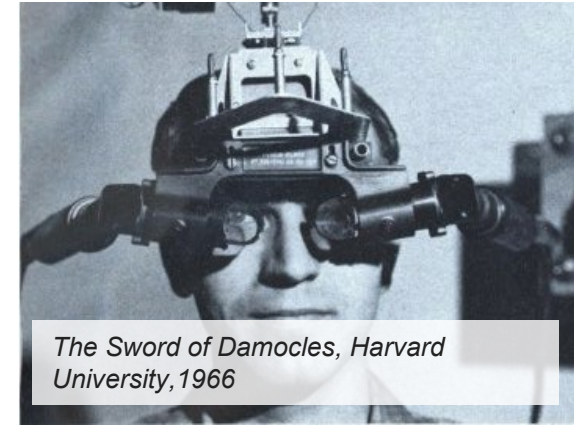


Interaction with and manipulation of both the physical and virtual environment.

Infographic Intel Lab

Spatial Computing

“use of space around us as a medium to interact with technology”



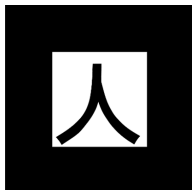
The Sword of Damocles, Harvard University, 1966



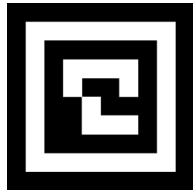
Microsoft, HoloLens 2, 2019

Situated Visualisation

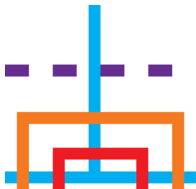
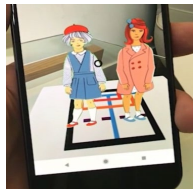
Marker based



ARToolkit



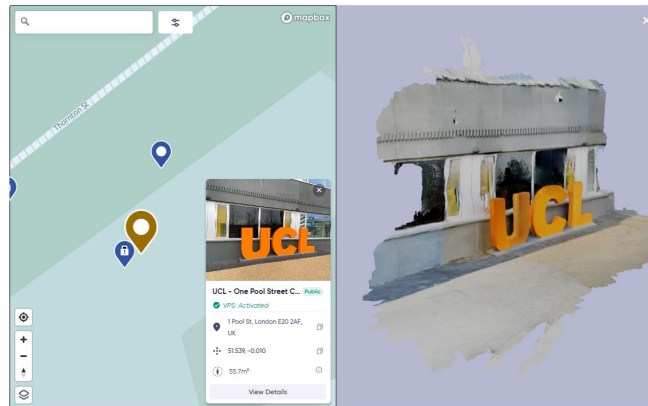
QRCode



Images



3D Object or Area (PC)



Niantic VPS



ARKit



ARCore



WebXR

NIANTIC

8thWall

Leah Lovett, Valerio Signorelli, Andy Hudson-Smith; Exploring the Materiality of Augmented Reality Markers through Arts-Led Cocreation: Drawing, Weaving, and Tiling. *Leonardo* 2024; 57 (4): 379–386. doi: https://doi.org/10.1162/leon_a_02542

Situated Visualisation

Markerless based



ARKit



ARCore



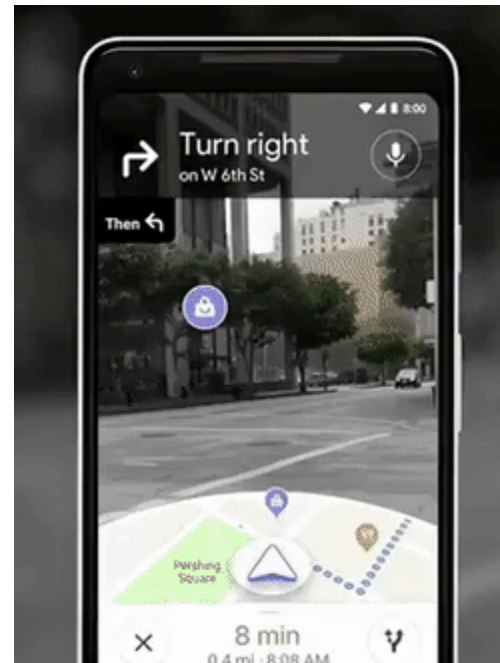
WebXR

NIANTIC

8thWall



Surface detection horizontal and vertical



Geolocation GPS



Geolocation WPS

Markerless based

Flat, Textureless Surfaces

Plain white desks or smooth walls offer little visual data, making it difficult to detect and anchor content.

Dim Lighting Conditions

Low light reduces the camera ability to detect surface features, leading to poor tracking performance.

Extremely Bright Environments

Overexposure can wash out visual details, making it hard for to distinguish surfaces and features.

Transparent or Reflective Surfaces

Glass, mirrors, or shiny floors can confuse

Dynamic or Moving Surfaces

Surfaces like rippling water, waving grass, or moving crowds are unstable and unreliable for anchoring AR content.

GPS Limitations

Tall buildings can block or reflect GPS signals,
It is generally unavailable or highly inaccurate indoors. Updates are relatively slow

VPS (Visual Positioning System)

VPS only works in locations that have been previously scanned and indexed. Changes in lighting or weather can affect the system's ability to match visual features. Temporary changes in the environment can interfere with recognition.

Situated Visualisation



MARS (“Mobile Augmented Reality System”) 1996. Columbia University
Computer Graphics and User Interfaces Lab
<http://graphics.cs.columbia.edu/projects/mars/>

T. Höllerer, S. Feiner, T. Terauchi, G. Rashid, D. Hallaway, Exploring MARS: Developing Indoor and Outdoor User Interfaces to a Mobile Augmented Reality System, *Computers and Graphics*, 23(6), Elsevier Publishers, Dec. 1999, 779-785

Object recognition

CARED

Connected Augmented Reality Energy Displays

```
▼ 107
▼ EM
  ▼ gosund
    ▼ peter-the-prusa-1
      LWT = Online
      STATE = {"Time":"2023-05-05T20:13:49","Uptime":"2T08:29:34","UptimeSec":203374,"Heap":27,"SleepMode":"Dynamic","Sleep":50,"Loa
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    ▼ pertuina-the-prusa-2
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      SENSOR = {"Time":"2023-05-05T20:14:03","ENERGY":{"TotalStartTime":"2022-11-16T16:28:08","Total":12.941,"Yesterday":0.024,"Today":
    ▼ sally-big-screen-logi1
      LWT = Online
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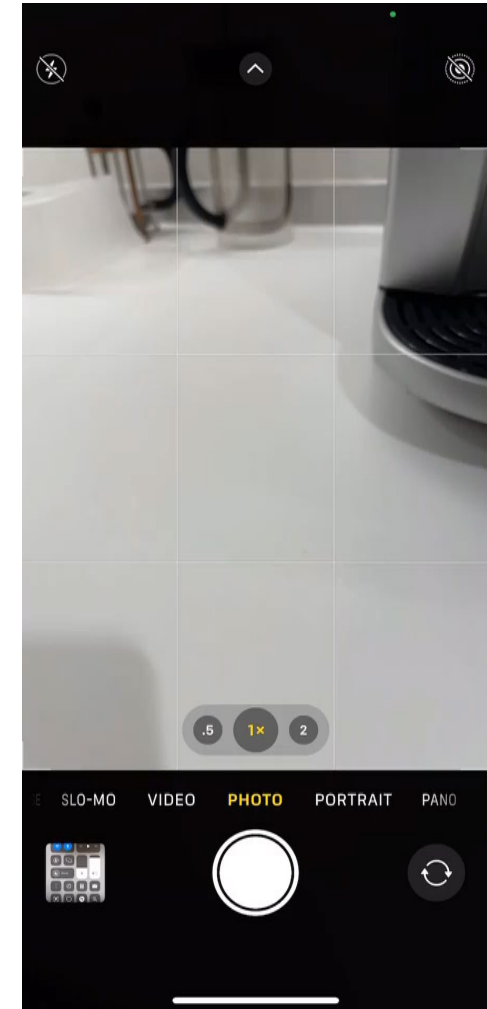


Image recognition

GROUNDbreakers *Immersive Trail & Guide*

Clarnico

The only remaining Victorian factory building in the Park provides us with a window into the production of sweet confectionary which were produced in Hackney Wick. The canals, railways and docks brought sugar and raw materials from abroad and allowed the finished sweets to be distributed around the world.

To find more Augmented Reality locations, visit the Google Map at livingmaps.org/art

telling the back story of the Queen Elizabeth Olympic Park site from the Bronze Age to the Digital Age

www.livingmaps.org/groundbreakers



**Scan the code to
step into history**

LIVINGMAPS

Heritage
5G

FOUNDATION
FOR FUTURE
LONDON

QUEEN
ELIZABETH
OLYMPIC PARK

HyperActive

Made possible with
Heritage
Fund

GroundBreakers

Telling the back story of the Queen Elizabeth Olympic Park site from Bronze Age to the Digital Age



Geofencing



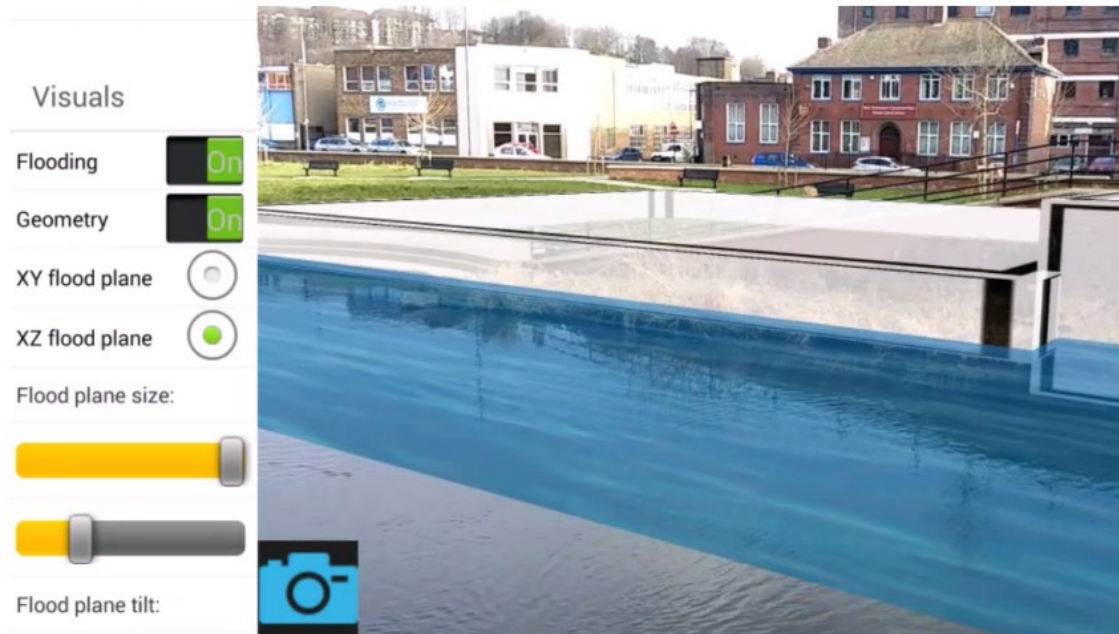
The Deep Listener.

Jakob Kudsk Steensen, Serpentine Galleries
Audio-visual walk through Hyde Park featuring five species: trees, bats, parakeets, damselflies and reedbeds.

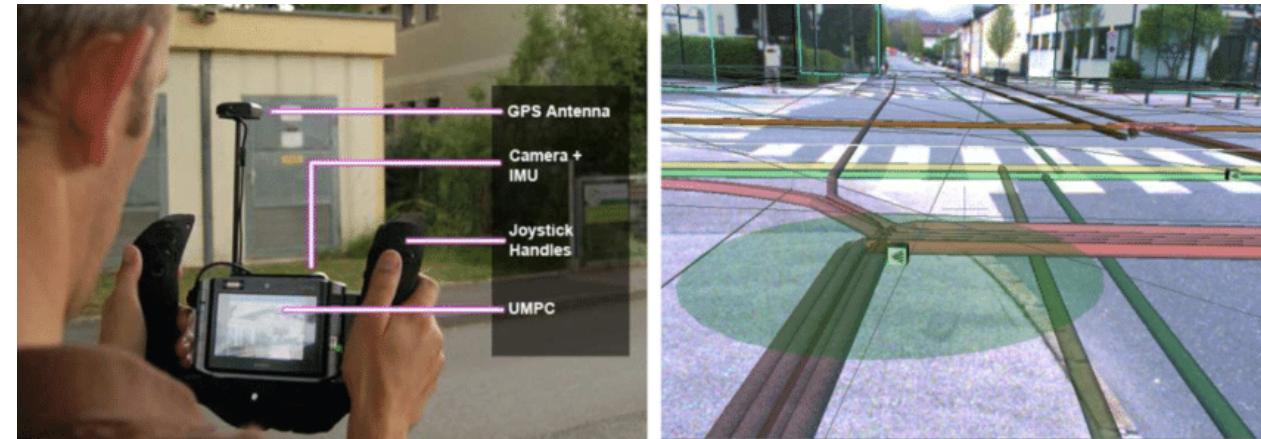


<https://augmentedarchitecture.org>

Geolocation



Haynes, P., Hehl-Lange, S., Lange, E. (2018) Mobile Augmented Reality for Flood Visualisation. *Environmental Modelling & Software* 109, Pages 380-389. <https://doi.org/10.1016/j.envsoft.2018.05.012>



Schall, G., Junghanns, S., & Schmalstieg, D. (2010). Vidente-3D Visualization of Underground Infrastructure using Handheld Augmented Reality. In *Geohydroinformatics : Integrating GIS and Water Engineering* CRC Press.

Other senses



AR soundscape app for train travellers
Alexander Pospischil

<https://www.alexanderpospischil.de/insideOutside.php>

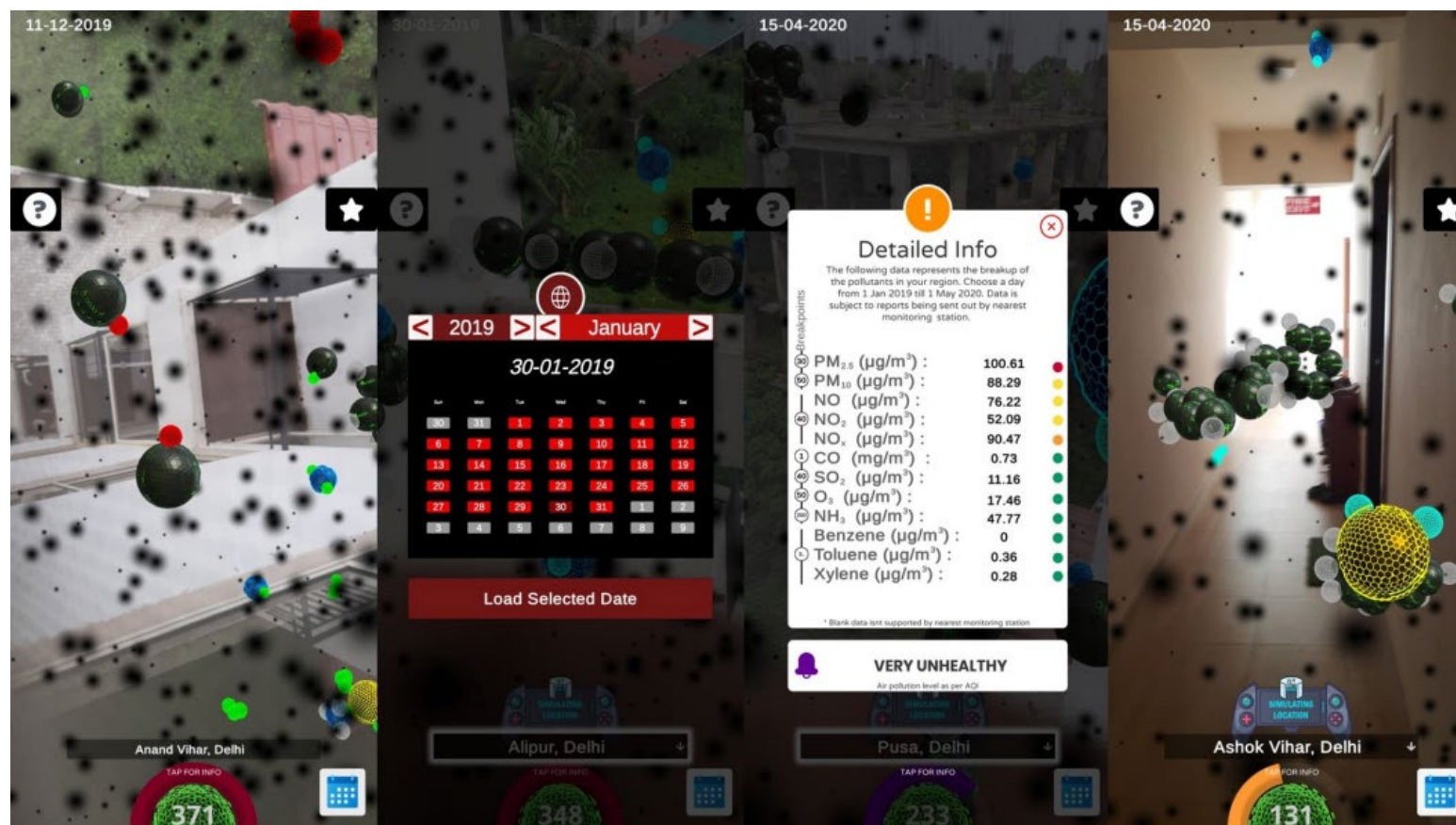


Microsoft Soundscape – Microsoft Research

explores the use of innovative audio-based technology to enable people to build a richer awareness of their surroundings, thus becoming more confident and empowered to get around.

Situated Visualisation

And extra-senses

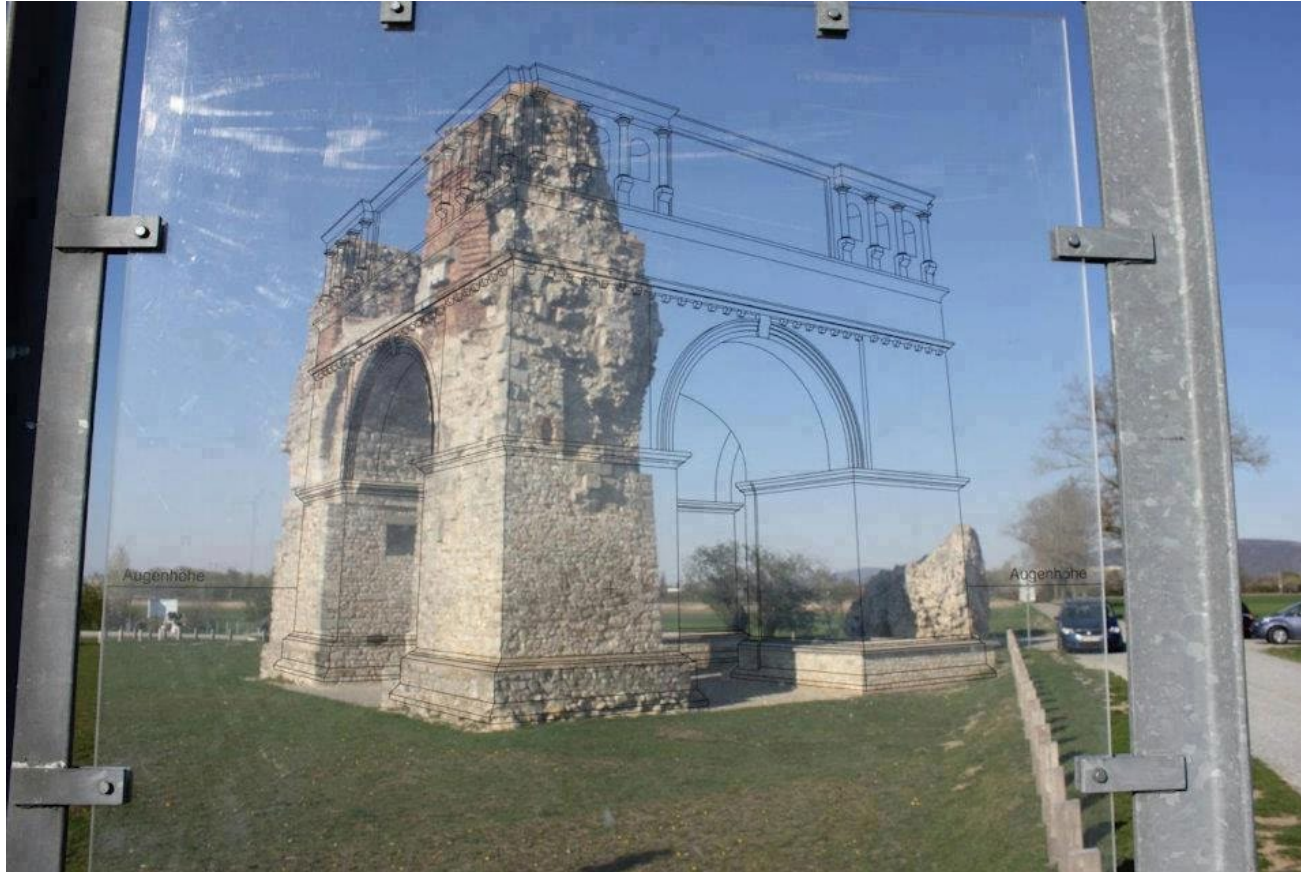


Mathews, N. S., Chimalakonda, S., & Jain, S. 2020. AiR, An Augmented Reality Application for Visualizing Air Pollution. <http://arxiv.org/abs/2006.02136>



Jansen, Y. et al., 2022. *Envisioning Situated Visualizations of Environmental Footprints in an Urban Environment*. [Doi:10.5281/zenodo.7053934](https://doi.org/10.5281/zenodo.7053934)

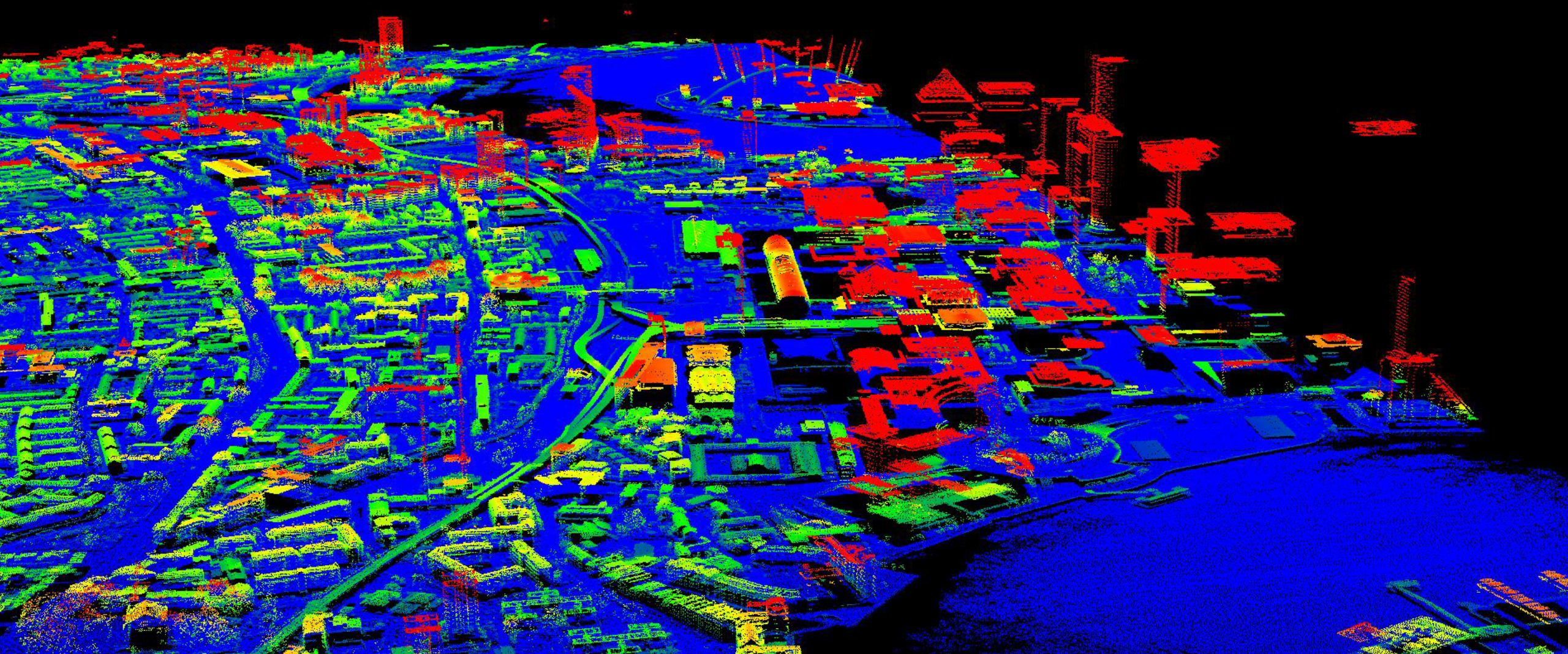
Physical display



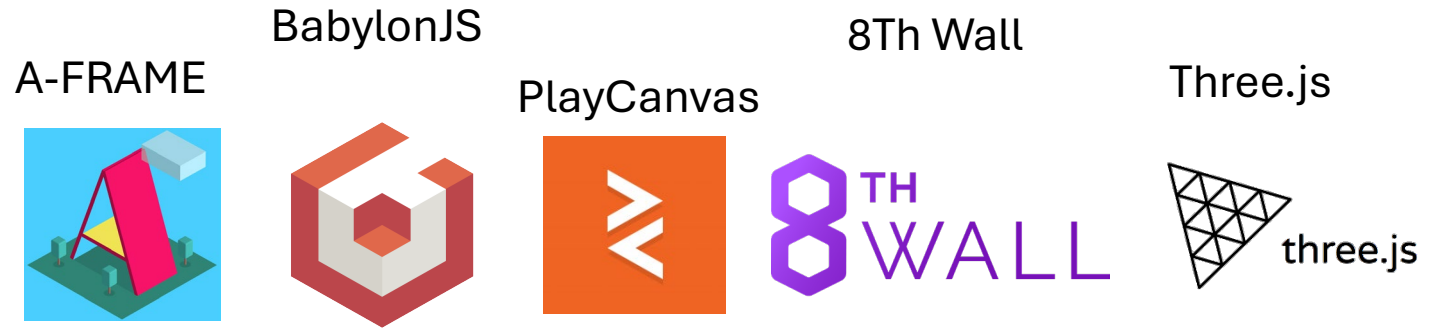
Heathen's Gate at Carnuntum, shown with an overlaid reconstruction drawing aligning the surviving Roman remains with their original architectural form.

- **Physical, site-integrated visualisation:**
Using fixed transparent panels, plaques, or installations to embed data directly into the environment, aligning information with the physical context of the site.
- **Augmentation without devices:**
Providing layered interpretation through tangible displays that add information the object or place cannot show on its own, enabling situated understanding without AR hardware.

Show me how



Frameworks and tools



Game Engine (Unity - Unreal) can export WebXR but not natively with some overhead.

HTML and JavaScript or TypeScript

Platform and Device Agnostic

Android | iOS | Win | Linux | MacOS

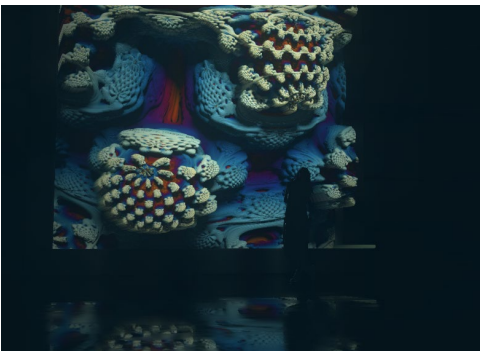
Meta Quest | HTC Vive | Hololens

Advantages

- Based on JS and HTML5 (open standards)
- Lightweight workflow: compatible with dev tools, version control, and rapid iteration
- Easy updates: in browser, no installation, no app store
- Cross-platform: one build runs on many devices
- Integrates with standard web APIs (WebGPU, WebAudio)

Disadvantages

- Performance limits: heavy scenes require optimisation
- Device diversity complicates testing
- Network-dependent: large assets → slow loads; offline requires PWA.
- Limited access to device-specific features (depends on browser support).
- Debugging can be harder compared to native tools



cineshader.com



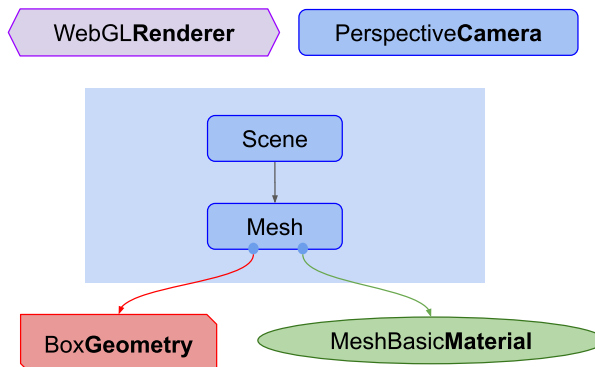
Low Poly Environment





HTML

```
<html lang="en" dir="ltr">
  <head>
    <title>Three.js Hello World!</title>
    <style>
      body { margin: 0; }
      canvas { width: 100%; height: 100% }
    </style>
  <!-- Three.js local -->
  <script src="js/three.min.js"></script>
</head>
<body>
  <script src="js/example.js"></script>
</body>
</html>
```



JavaScript

```
var scene = new THREE.Scene();
var camera = new THREE.PerspectiveCamera( 75, 1, 0.1, 1000 );
camera.position.z = 4;

var renderer = new THREE.WebGLRenderer({ antialias: true });
renderer.setSize( window.innerWidth, window.innerHeight );

document.body.appendChild( renderer.domElement );

var geometry = new THREE.BoxGeometry( 1, 1, 1 );
var material = new THREE.MeshBasicMaterial( { color: "#433F81" } );
var cube = new THREE.Mesh( geometry, material );

scene.add( cube );

renderer.render(scene, camera);
```

<https://threejsfundamentals.org/>



Frameworks and tools



Abir Holdings Inc is responsible for the deployment of technologies owned by Rok Venture Holdings LLC, a joint venture between world renowned Scientists and Business minds. We took on the challenge to make our planet cleaner, more efficient and technologically advanced by deploying impactful innovations.

We created, tested and patented a highly profitable Thermo-Chemical Method (TCM) system for treatment of municipal solid waste (MSW), along with commercial and industrial waste. Our TCM and Gasification System is sustainable, inexpensive, cost-effective and efficient. The Technology is free of pollution, ash or secondhand waste production. The intended system makes for cost effective power production as well as the creation of useful byproducts from any kind of raw material including hazardous waste.

Our mission is to enable Global Leaders to answer the most challenging question:

"What Should We Do With Our Wastefull?"

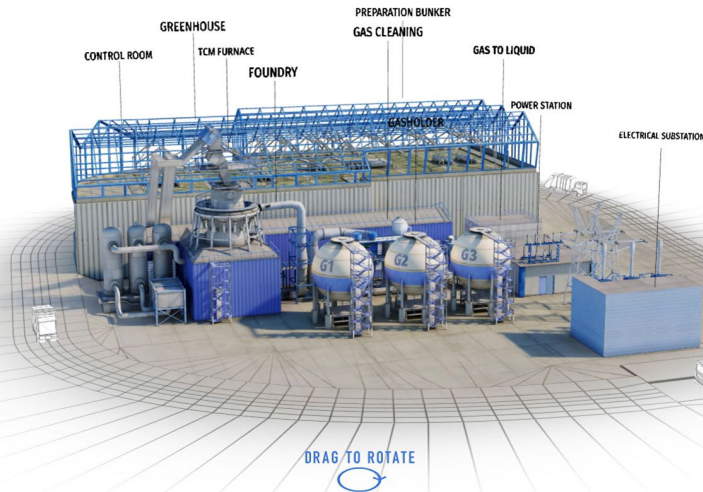
WASTE TO ENERGY TECHNOLOGY

⁰¹Scientific scheme / ⁰²Video explanation / ⁰³Related patent /

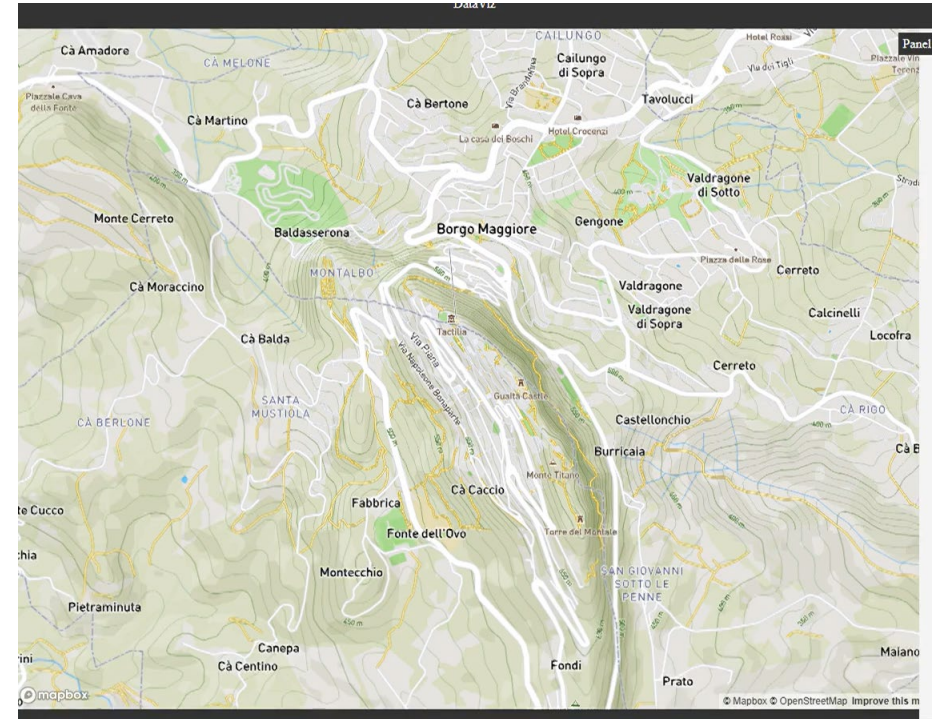
Over 2,000

landfills are currently in existence in the USA

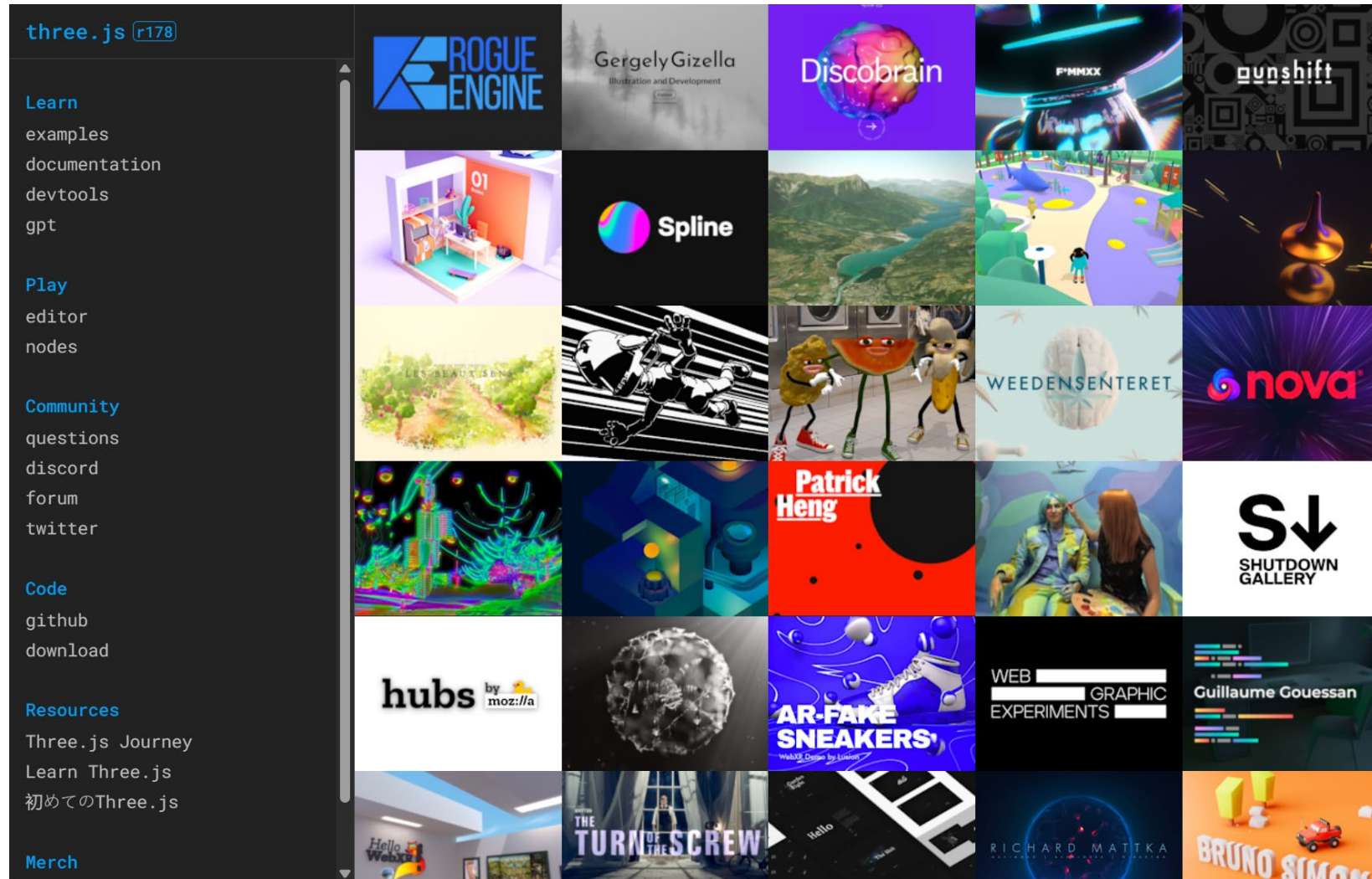
click on callouts to explore



ABIR – Energy Facility - BabylonJS

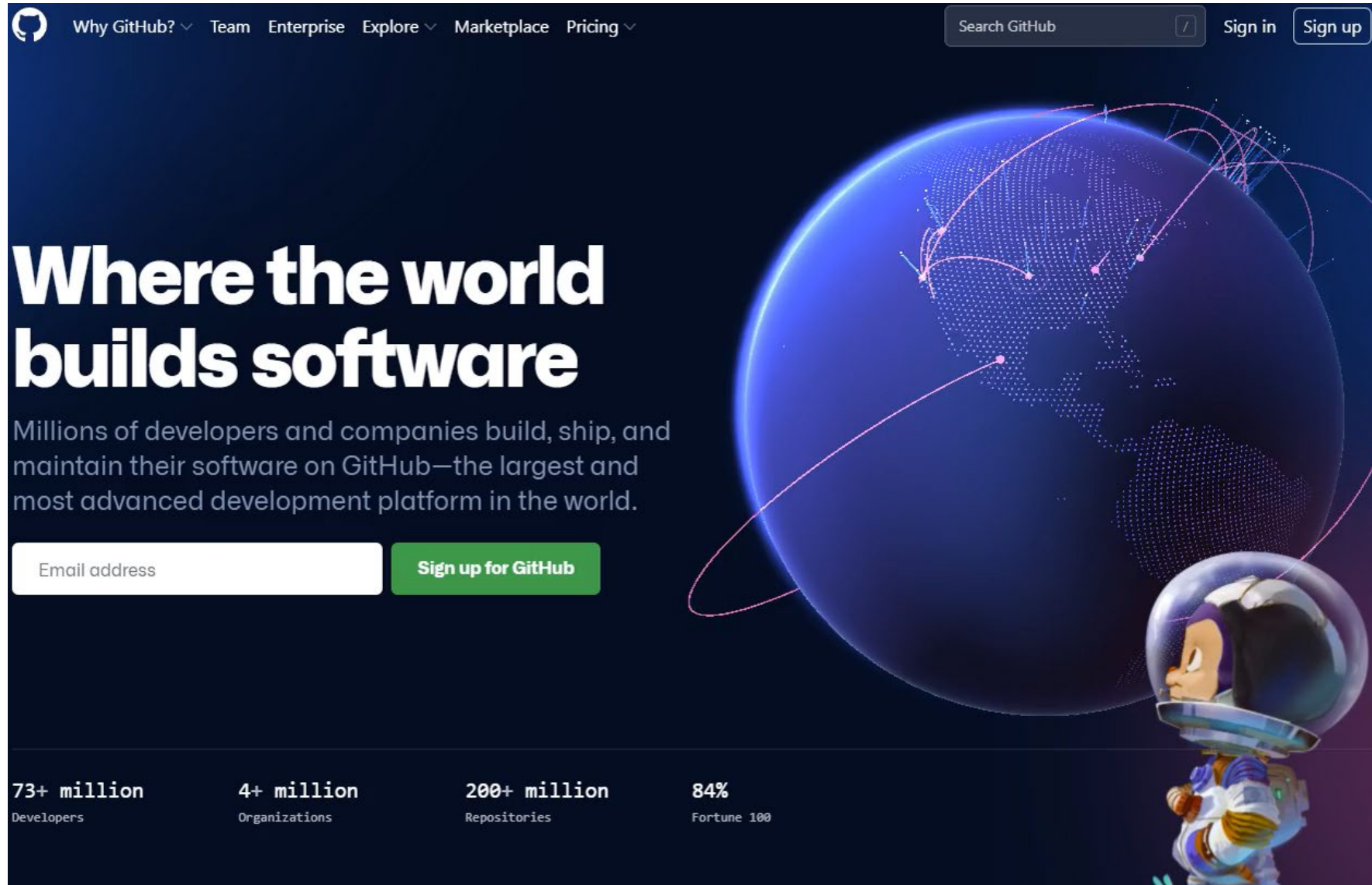


SanMarino – CesiumJS and Google 3Dtiles



A cross-browser JS library and API used to create and display animated 3D computer graphics in a web browser using WebGL

- Combined with other tools and frameworks (e.g. React, Angular, Svelte, Vue, Flutter)
- Data viz, ads, animation, videogame, storytelling
- 2D-3D interactive, Virtual Reality and Augmented Reality



The screenshot shows the GitHub homepage with a dark blue background. At the top, there's a navigation bar with links like 'Why GitHub?', 'Team', 'Enterprise', 'Explore', 'Marketplace', and 'Pricing'. A search bar and 'Sign in'/'Sign up' buttons are also present. The main headline reads 'Where the world builds software'. Below it, a sub-headline states 'Millions of developers and companies build, ship, and maintain their software on GitHub—the largest and most advanced development platform in the world.' A sign-up form with an 'Email address' input and a 'Sign up for GitHub' button is visible. On the right, there's a large graphic of a globe with glowing red lines representing code connections, and a small astronaut character in the bottom right corner. At the bottom, four statistics are listed: '73+ million Developers', '4+ million Organizations', '200+ million Repositories', and '84% Fortune 100'.

Why GitHub? Team Enterprise Explore Marketplace Pricing

Search GitHub Sign in Sign up

Where the world builds software

Millions of developers and companies build, ship, and maintain their software on GitHub—the largest and most advanced development platform in the world.

Email address Sign up for GitHub

73+ million Developers 4+ million Organizations 200+ million Repositories 84% Fortune 100

- *Standalone or Embedded with seamless interactivity*
- *Globe.GL for Data Visualisation*
 - *Real time pull requests*
- <https://github.blog/2020-12-21-how-we-built-the-github-globe/>

<https://github.com/home> [2022]

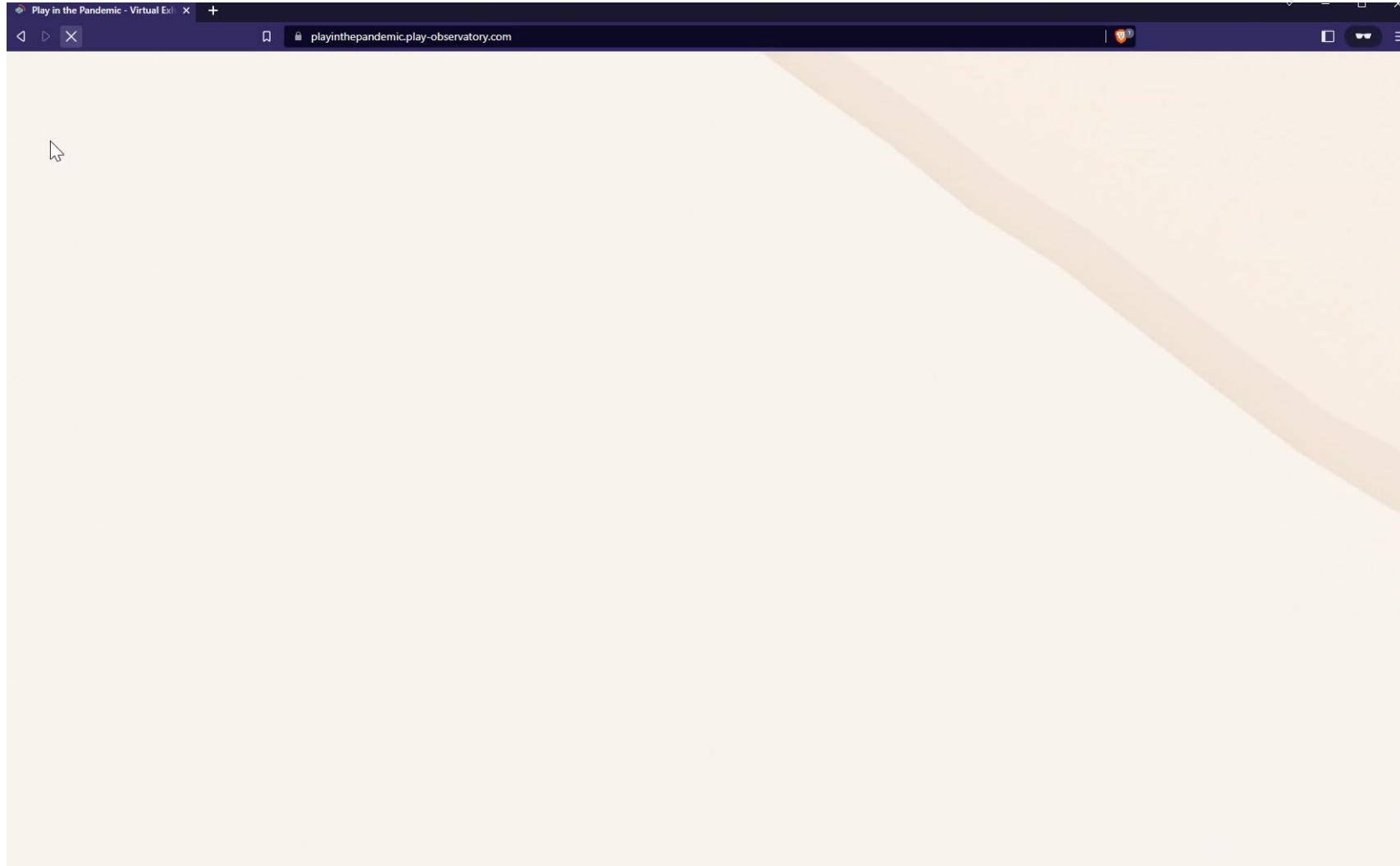


<https://demos.littleworkshop.fr/infinitytown>

- *Interoperability (tools)*
- *Procedural visualisation*
- *High performance also on mobile devices*

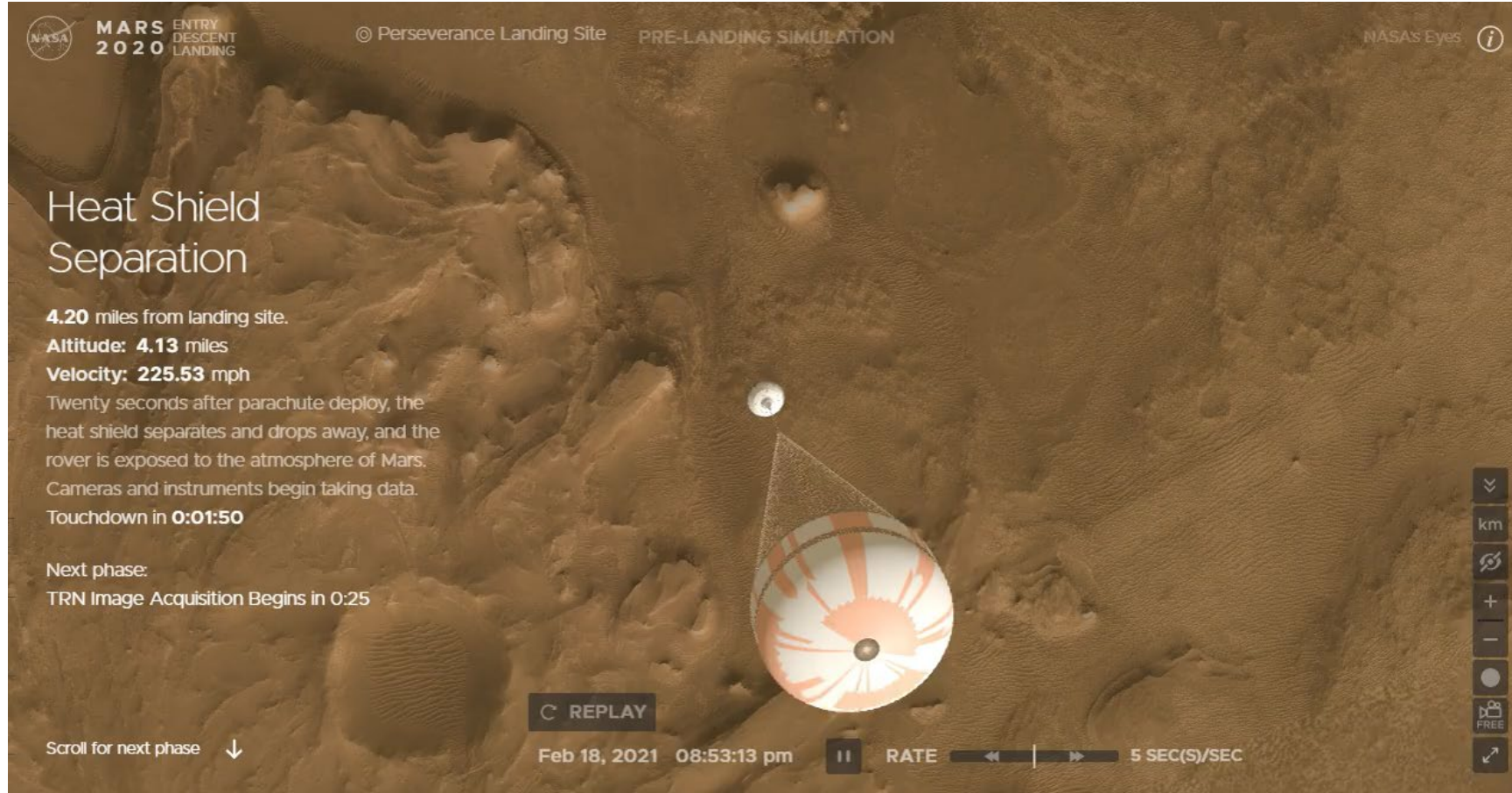


<https://slowroads.io/>



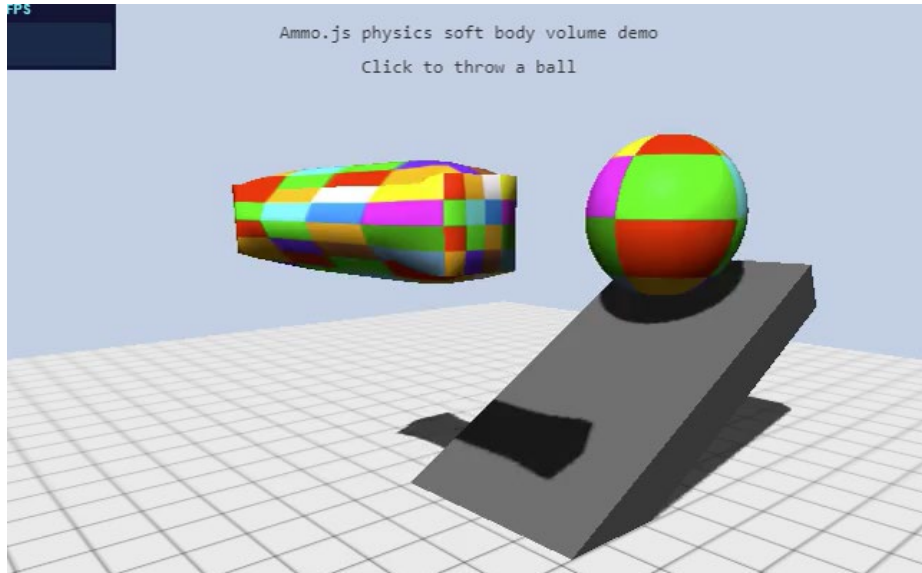
<https://playinthepandemic.play-observatory.com>

- *Storytelling and user engagement (play with website)*
- *Multidisciplinary*
 - *Coding – Web-dev*
 - *Graphic design*
 - *3D modelling*
 - *Data collection and archive*



- *Perseverance Mars Landing simulation*
- *Storytelling and data visualisation*
- *Realtime during the event*

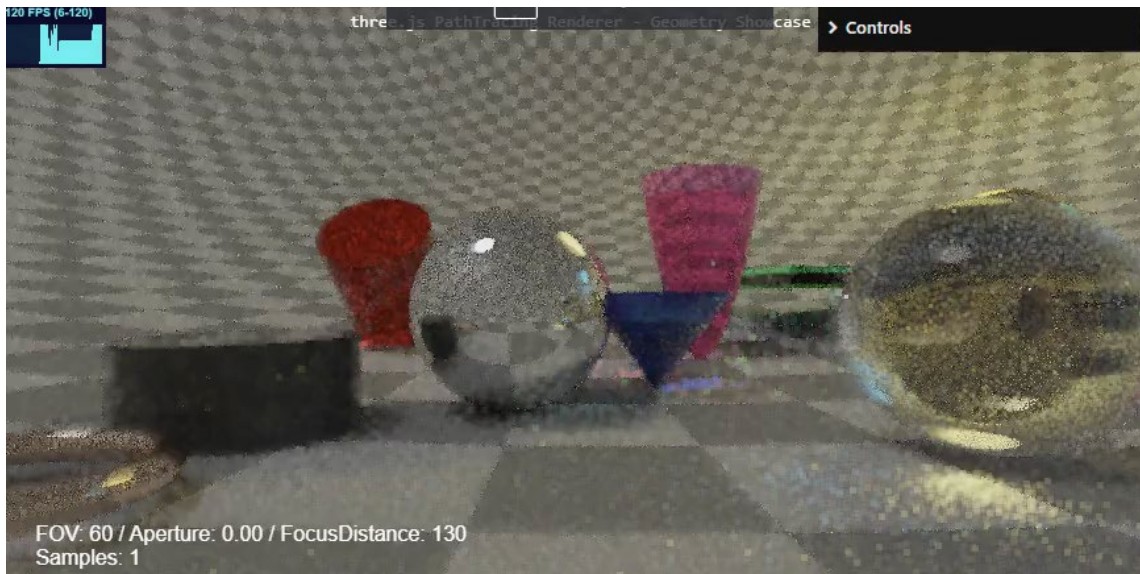
<https://eyes.nasa.gov/apps/mars2020/>



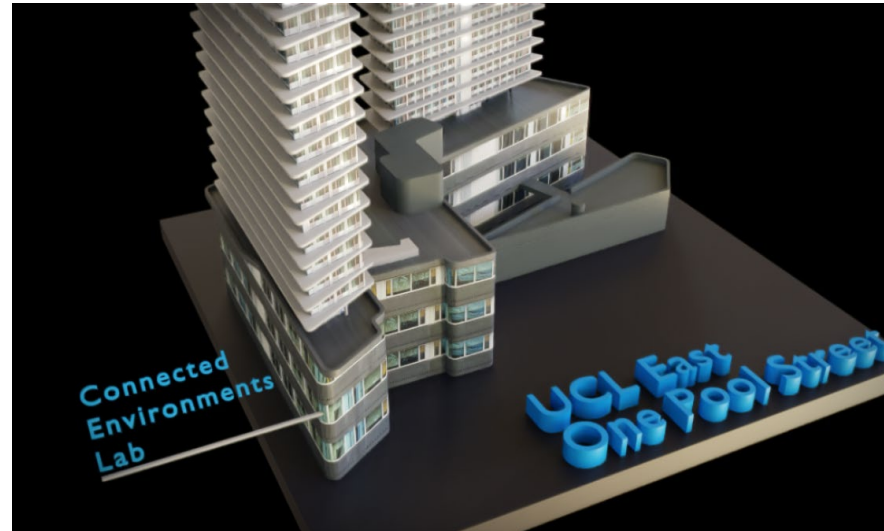
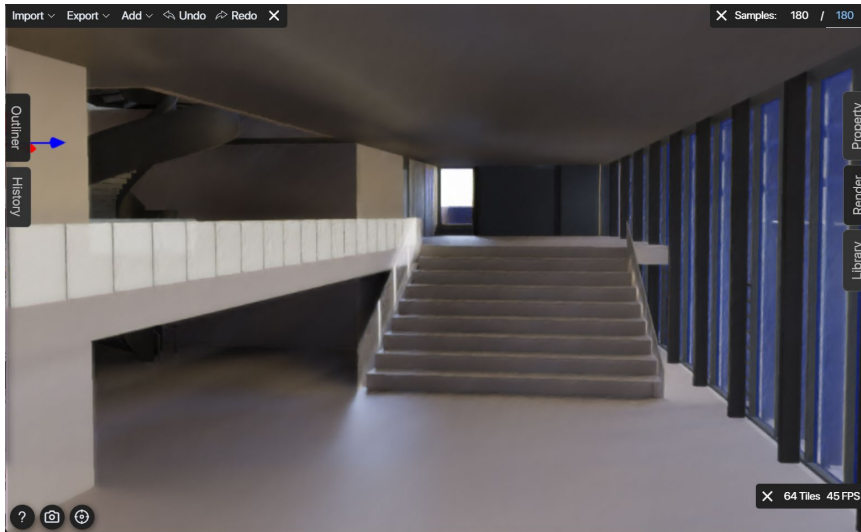
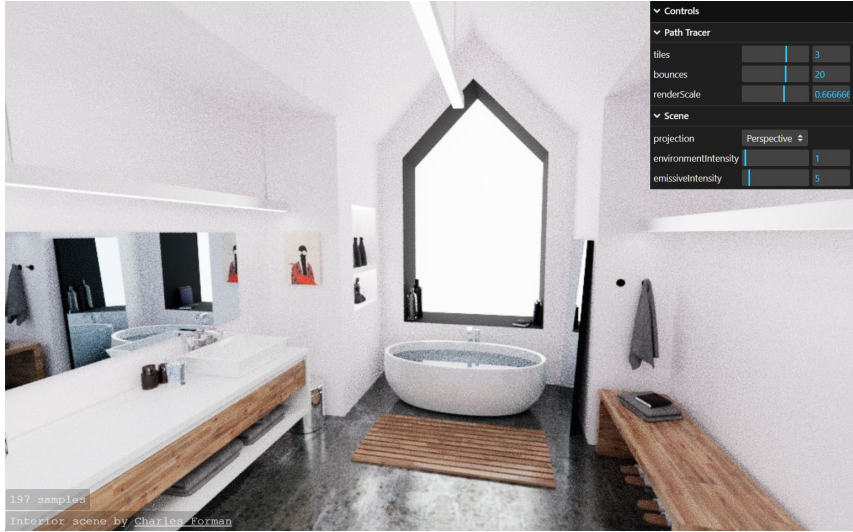
https://threejs.org/examples/#physics_ammo_volume

- *High performance*
- *PathTracing*
- *Real time physics simulation*
- *Entire interactive websites*

https://erichlof.github.io/THREE.js-PathTracing-Renderer/Geometry_Showcase.html

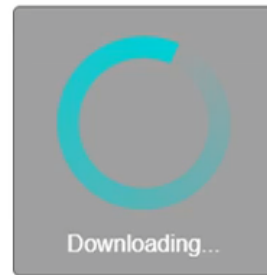


Frameworks and tools



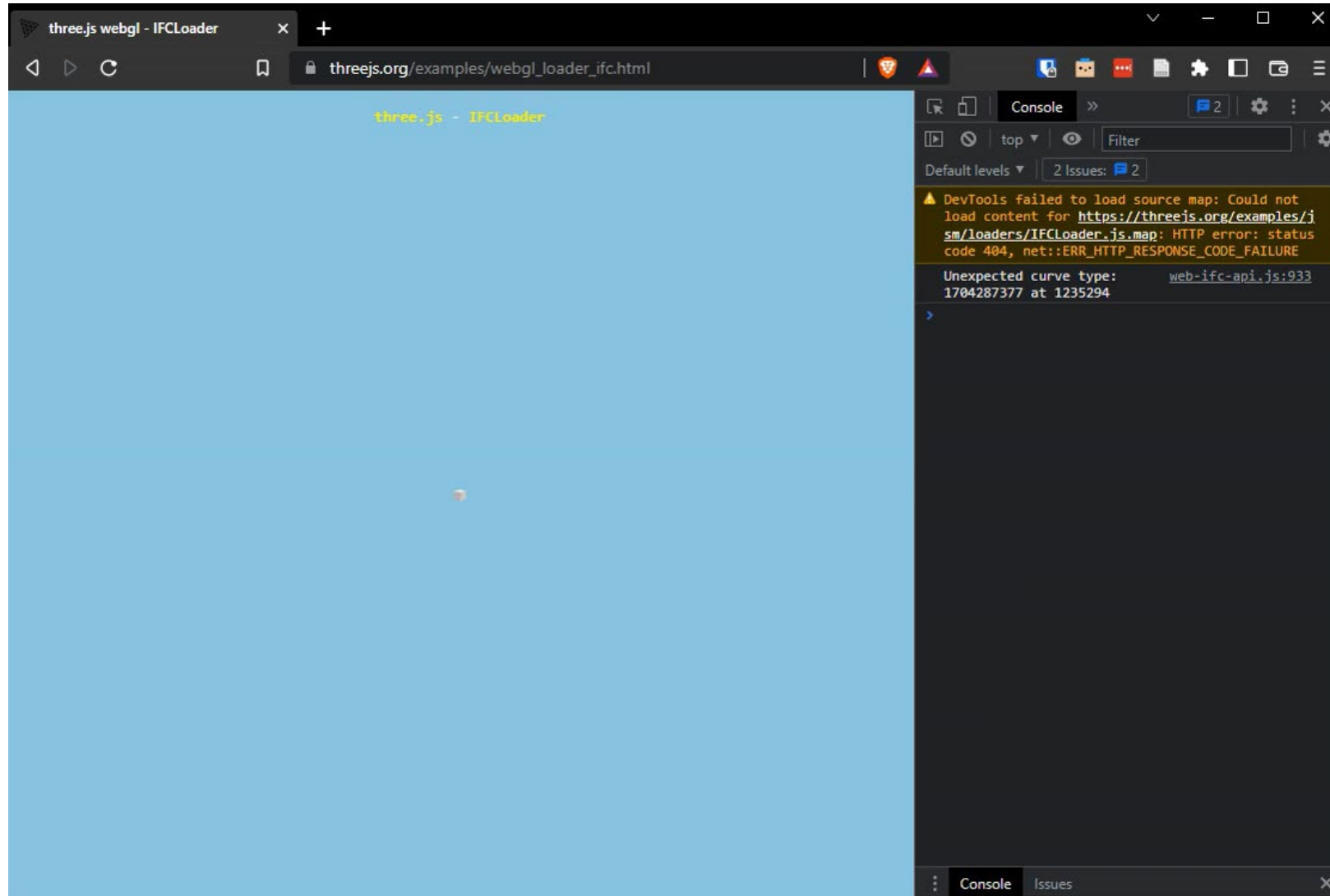
LGLTracer





- *Many built-in Loaders*
 - *Collada; FBX; GLTF; KMZ; OBJ; PLY; STL; TIFF; XYZ; USDZ; SVG*
- *IFC Loader*
- *URDF Loader*
- *3D Tiles Loader*
- *Gaussian Splatting*

https://projects.markkellogg.org/threejs/demo_gaussian_splats_3d.php



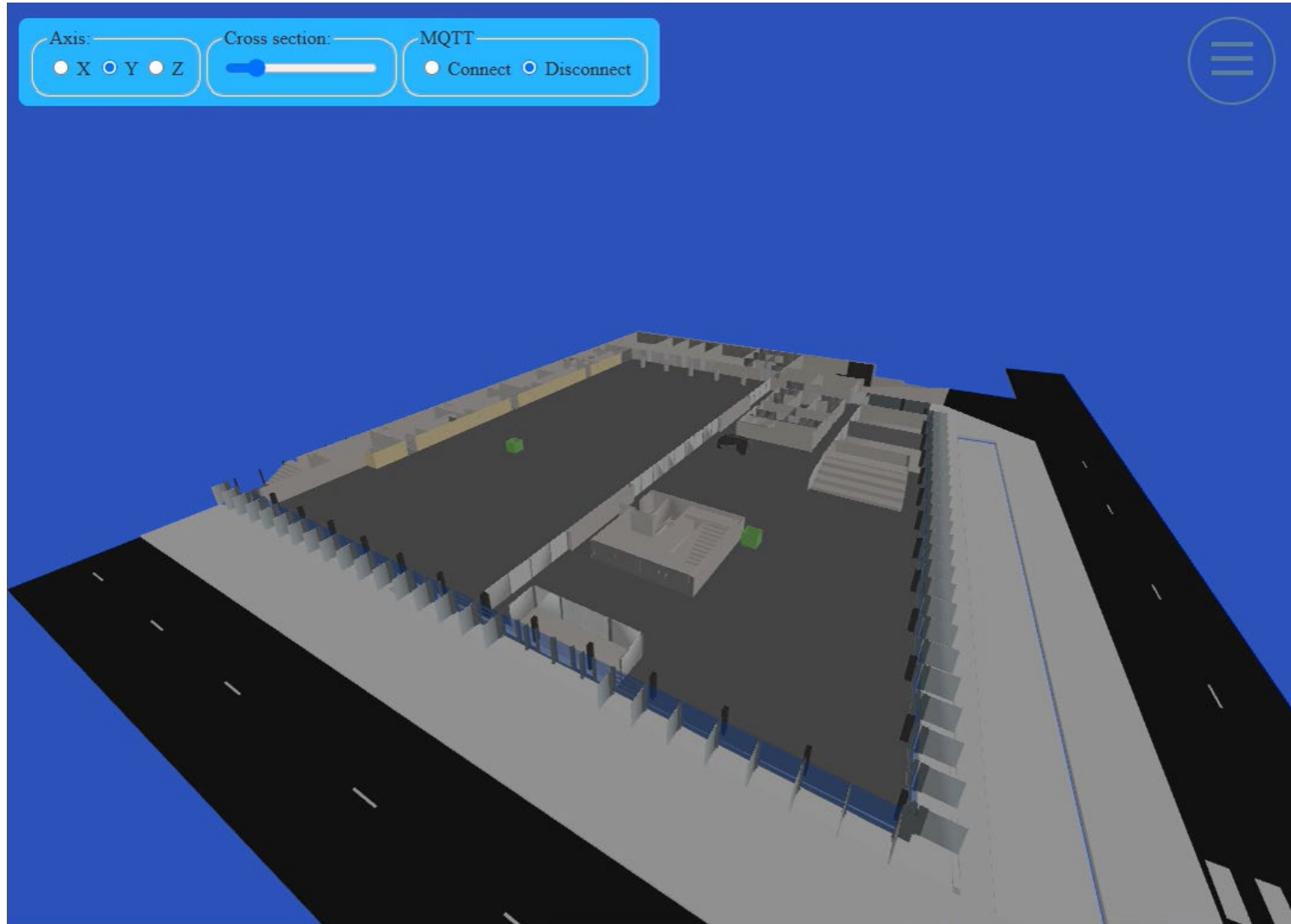
https://threejs.org/examples/webgl_loader_ifc.html

- *Many built-in Loaders*
 - *Collada; FBX; GLTF; KMZ; OBJ; PLY; STL; TIFF; XYZ; USDZ; SVG*
- *IFC Loader*
- *URDF Loader*
- *3D Tiles Loader*
- *Gaussian Splatting*



- *Many built-in Loaders*
 - Collada; FBX; GLTF; KMZ; OBJ; PLY; STL; TIFF; XYZ; USDZ; SVG
- *IFC Loader*
- *URDF Loader*
- *3D Tiles Loader*
- *Gaussian Splatting*

<https://gardenlab.cetools.org/>



- *Many built-in Loaders*
 - *Collada; FBX; GLTF; KMZ; OBJ; PLY; STL; TIFF; XYZ; USDZ; SVG*
- *IFC Loader*
- *URDF Loader*
- *3D Tiles Loader*
- *Gaussian Splatting*

IMREDD Nice – IFC and real time data